

Definition 7. We call **Banach space** a normed space which is **complete**, that is all its Cauchy sequences are convergent to an element of the space itself.

Exercise 3. Prove that $\mathbb{R}^n$ endowed with (1) is a Banach space, while $C_0^\infty(\mathbb{R})$ endowed with (2) is not, no matter the choice of p .

This last exercise raises naturally the question whether, given a space which is normed but not Banach, it is possible to complete it by adding a sufficient number of elements to make every Cauchy sequence convergent. The answer is positive and we omit the proof which is very similar to the one used to completing $\mathbb{Q}$ to $\mathbb{R}$ – see [5, Th. 2.15]:

Theorem 8. Let $(V, \|\cdot\|)$ be a normed space. Then

1. There exists a Banach space $\mathbb{B}(V)$, called **completion** of V such that V is isometric to a dense subspace³ of $\mathbb{B}(V)$ under a linear injective map $J : V \rightarrow \mathbb{B}(V)$,
2. if $(J_1, \mathbb{B}_1, \|\cdot\|_1)$ is a triple such that $(\mathbb{B}_1, \|\cdot\|_1)$ is a Banach space while $J_1 : V \rightarrow \mathbb{B}_1$ is a linear isometry whose image is dense in $\mathbb{B}_1$, then there exists a unique isomorphism of normed spaces $U : \mathbb{B}(V) \rightarrow \mathbb{B}_1$ such that $J_1 = U \circ J$.

1.2 Hilbert Spaces

While the notion of norm generalizes the idea of assigning a length to each element of a vector space and it leads to the notion of Banach space, we are still falling short of one key structure, namely we need to generalize the notion of angle between two vectors. In the following all vector spaces that we shall consider are assumed to be complex.

Definition 9. Let V be a complex vector space. We call **Hermitian inner product** (or *Hermitian scalar product*) a map $S : V \times V \rightarrow \mathbb{C}$ such that

1. $S(v, v) \geq 0$ for all $v \in V$,
2. $S(v, \alpha v' + \beta w) = \alpha S(v, v') + \beta S(v, w)$ for all $v, v', w \in V$ and for all $\alpha, \beta \in \mathbb{C}$,
3. $S(v, w) = \overline{S(w, v)}$ for all $v, w \in V$,
4. $S(v, v) = 0$ if and only if $v = 0$

Notice that the map S is often indicated simply as $(\cdot, \cdot)$, although we try to avoid this nomenclature at this stage. The following theorem characterizes a few notable properties of inner products, which are of paramount relevance to answer rather natural questions such as whether an inner product also identifies a norm.

³The notion of a dense set is developed in the Appendix, see Definition 215.

Theorem 10. *Let (V, S) be a vector space endowed with an Hermitian inner product. The Cauchy-Schwarz inequality holds:*

$$|S(v, w)|^2 \leq S(v, v)S(w, w), \quad \forall v, w \in V,$$

where the equality applies if and only if v and w are linearly dependent.

Proof. Let $v, w \in V$ be different from 0 and let $\lambda \in \mathbb{C}$. In view of 1. and 3. in Definition 9, it holds

$$0 \leq S(v + \lambda w, v + \lambda w) = S(v, v) + 2\Re(\lambda S(v, w)) + |\lambda|^2 S(w, w),$$

where in the equality we used also that S is bilinear. Choose $\lambda = -\frac{\overline{S(v, w)}}{S(w, w)}$ and the above inequality becomes

$$0 \leq S(v, v) - \frac{|S(v, w)|^2}{S(w, w)},$$

which entails the sought inequality. If either v or w vanish, the Cauchy-Schwarz inequality becomes trivial. Suppose now that v is proportional to w , say $w = \lambda v$, $\lambda \in \mathbb{C}$. Then the equality holds since

$$|S(v, w)|^2 = |\lambda|^2 S(v, v)^2 = S(v, v)S(\lambda v, \lambda v) = S(v, v)S(w, w).$$

Conversely, assume that there exists $w, v \in V$, both non vanishing, such that $|S(v, w)|^2 = S(v, v)S(w, w)$. Then, if we introduce $v' = \frac{v}{\sqrt{S(v, v)}}$ and $w' = \frac{w}{\sqrt{S(w, w)}}$, it holds that $|S(u', v')|^2 = 1$, hence $|S(u', v')| = 1$ being the absolute value of a complex number a non negative number. In other words, there exists $a \in \mathbb{R}$ such that $S(u', v') = e^{ia}$. We look for $\lambda, \mu \in \mathbb{C}$ such that $\lambda u' + \mu v' = 0$. By condition 4. in Definition 9, this is tantamount to claiming that

$$0 = S(\lambda u' + \mu v', \lambda u' + \mu v') = |\lambda|^2 + |\mu|^2 + 2\Re(\bar{\lambda}\mu e^{ia}),$$

where we used that $S(u', u') = S(v', v') = 1$ per construction. It suffices to choose $\lambda = e^{ib}$, $\mu = e^{ic}$ with $-b + c + a = \pi$ and we have found the sought constants for which v' and u' and, consequently also u and v are proportional. $\square$

Exercise 4. Prove the Cauchy-Schwarz inequality for a vector space V endowed with an Hermitian semi-inner product, namely a map S which enjoys all properties of Definition 9 barring 4. (Suggestion: Follow the proof of Theorem 10, but notice that $S(w, w)$ can vanish even if $w \neq 0$.)

Thanks to Theorem 10, we are ready to spell out the following remark

Remark 1. *Every vector space with an Hermitian inner product (V, S) is also a normed space since one can define $\|\cdot\| : V \rightarrow \mathbb{R}$ as*

$$\|v\| = \sqrt{S(v, v)}, \quad \forall v \in V. \quad (3)$$

Comparing with Definition 1, we see that (3) meets automatically the requirements 1., 2. and 4. The only one which is not obvious is the third. Yet, if $v, w \in V$, it holds that

$$\begin{aligned}\|v + w\|^2 &= S(v + w, v + w) = S(v, v) + S(v, w) + S(w, v) + S(w, w) = \\ &= S(v, v) + S(w, w) + 2\Re(S(v, w)).\end{aligned}$$

Furthermore, in view of the Cauchy-Schwarz inequality $\Re(S(v, w)) \leq |S(v, w)| \leq \|v\| \|w\|$ which entails the triangle inequality $\|v + w\|^2 \leq \|v\|^2 + \|w\|^2 + 2\|v\| \|w\| = (\|v\| + \|w\|)^2$.

In addition, since every vector space with an Hermitian inner product is also a normed space, we can apply Definition 2 and 3 to endow it with norm topology.

Remark 2. A second notable feature of an Hermitian inner product S on a vector space V is the possibility of defining a notion of orthogonality between two vectors $u, v \in V$, which we indicate by $u \perp v$ and which is characterized by the equation $S(u, v) = 0$. More generally if $W \subseteq V$ is a vector subspace of V , we introduce the **orthogonal space** to W as

$$W^\perp \doteq \{v \in V \mid S(w, v) = 0, \forall w \in W\}.$$

Observe that, if we consider a third vector subspace $W' \subseteq W \subseteq V$, then $W^\perp \subseteq W'^\perp$.

Exercise 5. Let (V, S) be a vector space endowed with an Hermitian inner product. Show that

1. the **parallelogram rule** holds: For all $v, w \in V$

$$\|v + w\|^2 + \|v - w\|^2 = 2(\|v\|^2 + \|w\|^2), \quad (4)$$

where $\|v\|^2 = S(v, v)$ for all $v \in V$.

2. the **polarisation identity** holds: For all $v, w \in V$

$$S(v, w) = \frac{1}{4} (\|v + w\|^2 - \|v - w\|^2 - i\|v + iw\|^2 + i\|v - iw\|^2). \quad (5)$$

Since all vector spaces with an Hermitian inner product are also normed spaces, the notion of a continuous map between two such spaces is taken verbatim from Definition 4. Nonetheless the notion of isometry and of isomorphism need a few rearrangements:

Definition 11. Let (V, S) and (V', S') be two vector spaces endowed with an Hermitian inner product. A linear map $U : V \rightarrow V'$ is called an **isometry** if

$$S'(U(v), U(w)) = S(v, w), \quad \forall v, w \in V.$$

If U is moreover injective and surjective we call it an **isomorphism** and we say that (V, S) and (V', S') are isomorphic as inner product spaces under the action of U .

Exercise 6. Show that every isometry between two vector spaces endowed with an Hermitian inner product is necessarily injective.

Notice that a vector space endowed with an Hermitian inner product suffers of the same problem of a normed space, namely we can introduce a notion of convergence of sequences following Definition 6, but we have no guarantee a priori that all Cauchy sequences are converging to an element of the vector space. This leads to the following definition:

Definition 12. We call Hilbert space $\mathcal{H}$ a complex vector space equipped with an Hermitian inner product S which is a Banach space with respect to the norm induced by S .

Notice that an isomorphism between Hilbert spaces is also called a **unitary operator**, a nomenclature which we will be using henceforth. The following theorem characterizes better the relation between Definition 12 and that of a vector spaces with an Hermitian inner product:

Theorem 13. Let (V, S) be a complex vector space V endowed with an Hermitian inner product S . Then

1. there exist a Hilbert space $\mathcal{H}$ together with an Hermitian inner product $(\cdot, \cdot)$ as well as an injective map $J : V \rightarrow \mathcal{H}$ such that $J(V)$ is dense in $\mathcal{H}$ and $(J(v), J(v')) = S(v, v')$ for all $v, v' \in V$.
2. if there exists a second triple $(J_1, \mathcal{H}_1, (\cdot, \cdot)_1)$ such that $J_1 : V \rightarrow \mathcal{H}_1$ is an injective map whose image is dense in $\mathcal{H}_1$ and $(J_1(v), J_1(v'))_1 = S(v, v')$ for all $v, v' \in B$, then there exists a unique unitary operator $U_1 : \mathcal{H} \rightarrow \mathcal{H}_1$ such that $U_1 \circ J = J_1$.

1.3 Reflexivity of Hilbert Spaces

Goal of this subsection is to prove one of the most important properties of Hilbert spaces: They are *reflexive*, namely they are (canonically) isomorphic to their dual space exactly as every finite dimensional vector space. As we will see this is a by-product of Riesz representation theorem. We recall a basic definition of set theory:

Definition 14. Let V be a vector space over the complex numbers and let K be any non empty subset. We say that K is **convex**, if $\forall v, v' \in K$ and $\forall \lambda \in [0, 1]$, $\lambda v + (1 - \lambda)v' \in K$.

Following the notation of [5], we call $\langle K \rangle$ the vector subspace of V generated by all linear combinations of elements in K and we denote with $\overline{K}$ the closure of K in V , whenever V is endowed with a suitable notion of convergence of sequences, *e.g.*, when it has a topology.

Theorem 15. Let $\mathcal{H}$ be a Hilbert space and let $K \subseteq \mathcal{H}$ be a non empty set. Then

1. $K^\perp \doteq \{\psi \in \mathcal{H} \mid (\psi, \phi) = 0, \forall \phi \in K\}$ is a closed subspace of $\mathcal{H}$,
2. $K^\perp = \langle K \rangle^\perp = \overline{\langle K \rangle}^\perp = \overline{\langle K \rangle^\perp}$,
3. If $K \subseteq \mathcal{H}$ is a closed subspace, than $\mathcal{H} = K \oplus K^\perp$,
4. $(K^\perp)^\perp = \overline{\langle K \rangle}$.

Proof. 1. $K^\perp$ is a linear space on account of the bilinearity of the Hermitian inner product. Let $\{\psi_n\}_{n \in \mathbb{N}}$ be a Cauchy sequence of elements in $K^\perp \subseteq \mathcal{H}$. Since $\mathcal{H}$ is complete per hypothesis, there must exist $\psi \in \mathcal{H}$ such that $\lim_{n \rightarrow \infty} \psi_n = \psi$. At the same time, for all $n \in \mathbb{N}$, $(\psi_n, \phi) = 0$ for all $\phi \in K$ and, by the continuity of $(\cdot, \cdot)$ we can take the limit as $n \rightarrow \infty$, obtaining $(\psi, \phi) = 0$, that is $\psi \in K^\perp$.

2. The first identity descends trivially from the bilinear structure of the Hermitian inner product, while the second is nothing but item 1. The third equality follows from this reasoning: Since $\langle K \rangle \subseteq \overline{\langle K \rangle}$, then $\overline{\langle K \rangle}^\perp \subseteq \langle K \rangle^\perp$. At the same time, per definition of closure, it holds $\langle K \rangle^\perp \subseteq \overline{\langle K \rangle}^\perp$, from which it descends $\langle K \rangle^\perp = \overline{\langle K \rangle}^\perp$.

3. Since $K \cap K^\perp = \{0\}$ per construction, we need to show that $\langle K, K^\perp \rangle = \mathcal{H}$, that is $\mathcal{H}$ is generated by linear combinations of elements in K and $K^\perp$. Let $\psi \in \mathcal{H}$ and choose $\phi \in K$ so to minimize $\|\psi - \phi\|$ as $\phi \in K$ varies. Let $\tilde{\phi} \doteq \psi - \phi$. If we show that $\tilde{\phi} \in K^\perp$ we have concluded. Let $\phi_0 \in K$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be

$$f(t) \doteq \|\psi - \phi + t\phi_0\|^2.$$

We choose ϕ_0 so that f is extremal⁴ at $t = 0$. To this end, we compute

$$f'(0) = \lim_{\epsilon \rightarrow 0} \frac{\|\tilde{\phi} + \epsilon\phi_0\|^2 - \|\tilde{\phi}\|^2}{\epsilon} = 2\Re(\tilde{\phi}, \phi_0).$$

Setting $f'(0) = 0$, we obtain that the real part of the inner product between $\tilde{\phi}$ and ϕ_0 vanishes. Repeating the same calculation with $i\phi_0$ in place of ϕ_0 yields that also the imaginary part of the inner product is 0. In other words $\phi_0 \in K^\perp$.

4. Since $K \perp K^\perp$ per construction and since the Hermitian inner product on $\mathcal{H}$ is continuous, we can consider the completion of $\langle K \rangle$ obtaining that $\overline{\langle K \rangle} \subset (K^\perp)^\perp$. Using both the second and the third item of this theorem, we obtain

$$\mathcal{H} = \overline{\langle K \rangle} \oplus \overline{\langle K \rangle}^\perp = \overline{\langle K \rangle} \oplus K^\perp.$$

Suppose that $\psi \in (K^\perp)^\perp \subset \mathcal{H}$. There must exist $\phi_0 \in \overline{\langle K \rangle}$ and $\phi_1 \in K^\perp$ such that $\psi = \phi_0 + \phi_1$. Taking the scalar product with ϕ_1 , we obtain that

$$(\phi_1, \psi) = \|\phi_1\|^2 = 0,$$

where we used that $K^\perp \perp \overline{\langle K \rangle}$ as well as the hypothesis that $\psi \in (K^\perp)^\perp$. As a consequence of $\|\phi_1\| = 0$, $\phi_1 = 0$. Hence $\psi = \phi_0 \in \overline{\langle K \rangle}$, that is $(K^\perp)^\perp \subseteq \overline{\langle K \rangle}$, which concludes the proof. $\square$

Corollary 16. *Let $\mathcal{H}$ be a Hilbert space. A subset $K \subset \mathcal{H}$ is dense in $\mathcal{H}$ if and only if $K^\perp = \{0\}$.*

Proof. This is a straightforward consequence of item 2. and 4. of Theorem 15. $\square$

We can prove the main statement of this subsection:

Theorem 17 (Riesz). *Let $\mathcal{H}$ be a Hilbert space. For any continuous linear functional⁵ $F :$*

⁴Notice that this is actually a minimum.

⁵The notion of continuous functional is the one of Definition 4.

$\mathcal{H} \rightarrow \mathbb{C}$, there exists a unique $\psi_F \in \mathcal{H}$ such that

$$F(\psi) = (\psi_F, \psi).$$

The map $F \mapsto \psi_F$ is a bijection.

Proof. Let F be any continuous linear functional as per hypothesis. Continuity entails that $\ker(F) = \{\psi_0 \in \mathcal{H} \mid F(\psi_0) = 0\}$ is a closed subspace of $\mathcal{H}$. Theorem 15 guarantees us that $\mathcal{H} = \ker(F) \oplus \ker(F)^\perp$. If $\ker(F) = \mathcal{H}$, then we can choose $\psi_F = 0$. If not, then $\dim(\ker(F)^\perp) = 1$. To prove it, let $\phi \in \ker(F)^\perp$. For any $\phi_1 \in \ker(F)^\perp$, $\phi_1 - \frac{F(\phi_1)}{F(\phi)}\phi \in \ker(F)^\perp$ since $\ker(F)^\perp$ is a closed subspace. At the same time this linear combination lies in $\ker(F)$ since

$$F\left(\phi_1 - \frac{F(\phi_1)}{F(\phi)}\phi\right) = F(\phi) - \frac{F(\phi_1)}{F(\phi)}F(\phi) = 0.$$

Yet, we know that $\ker(F) \cap \ker(F)^\perp = \{0\}$. Hence $\phi_1 - \frac{F(\phi_1)}{F(\phi)}\phi = 0$, which means that ϕ generates the whole $\ker(F)^\perp$. We can now make the following ansatz:

$$\psi_F \doteq \frac{\overline{F(\phi)}}{\|\phi\|^2}\phi.$$

Let $\psi \in \mathcal{H}$. We can write it as $\psi = \tilde{\psi} + \tilde{\phi}$ where $\tilde{\psi} \in \ker(F)$ and $\tilde{\phi} \in \ker(F)^\perp$. The previous analysis tells us that $\tilde{\phi} = \frac{F(\tilde{\phi})}{F(\phi)}\phi$. Yet, per linearity of F , $F(\psi) = F(\tilde{\psi}) + F(\tilde{\phi}) = F(\tilde{\phi})$. Hence

$$\tilde{\phi} = \frac{F(\psi)}{F(\phi)}\phi, \implies \psi = \tilde{\psi} + \frac{F(\psi)}{F(\phi)}\phi$$

As a by-product

$$(\psi_F, \psi) = \left(\frac{\overline{F(\phi)}}{\|\phi\|^2}\phi, \tilde{\psi} + \frac{F(\psi)}{F(\phi)}\phi\right) = \left(\frac{\overline{F(\phi)}}{\|\phi\|^2}\phi, \frac{F(\psi)}{F(\phi)}\phi\right) = F(\psi),$$

where we used in the second equality that $(\tilde{\psi}, \phi) = 0$, while, in the third, that the Hermitian inner product is anti-linear in the first entry. The map $F \mapsto \psi_F$ is well-defined and also injective. This is consequence of the fact that the only continuous map F such that $F(\psi) = 0$ for all $\psi \in \mathcal{H}$ can only be the trivial map. $F \mapsto \psi_F$ is also surjective, since, for all $\psi' \in \mathcal{H}$, one can define a continuous linear functional $F_{\psi'} : \mathcal{H} \rightarrow \mathbb{C}$ as $F_{\psi'}(\psi) \doteq (\psi', \psi)$. $\square$

Corollary 18. Every Hilbert space $\mathcal{H}$ is reflexive, that is, if one considers $\mathcal{H}^* \doteq \{F : \mathcal{H} \rightarrow \mathbb{C} \mid F \text{ is continuous and linear}\}$, the linear map $\mathcal{I} : \mathcal{H} \rightarrow (\mathcal{H}^*)^*$, such that

$$\mathcal{I}(\psi)[F] = F(\psi), \quad \forall \psi \in \mathcal{H}, \quad \text{and} \quad \forall F \in \mathcal{H}^*$$

is a unitary operator, namely a surjective isometry.

Proof. Riesz theorem 17 entails the existence of an isomorphism between $\mathcal{H}$ and its dual $\mathcal{H}^*$. In addition, we can turn this map into an isometry of Hilbert spaces if we endow $\mathcal{H}^*$ with the Hermitian inner product

$$(F', F)_{\mathcal{H}^*} \doteq (\psi_F, \psi_{F'}), \quad (6)$$

where ψ_F and $\psi_{F'}$ are constructed from F and F' as per theorem 17. By applying once more Riesz theorem, we can infer that for every $H \in (\mathcal{H}^*)^*$, there must exist $F_H \in \mathcal{H}^*$ such that $H(F') = (F_H, F')_{\mathcal{H}^*}$ for all $F' \in \mathcal{H}^*$. On the one hand repeating the same line of reasoning as above, we can conclude that there exists an isometry between $\mathcal{H}^*$ and $\mathcal{H}^{**}$ if we endow the latter with the Hermitian inner product

$$(H, H')_{\mathcal{H}^{**}} \doteq (F_{H'}, F_H)_{\mathcal{H}^*}.$$

Combining these results, it descends that there exists an isometry of Hilbert spaces between $\mathcal{H}$ and $\mathcal{H}^{**}$. On the other hand, equation (6) entails that $(F_H, F')_{\mathcal{H}^*} = (\psi_{F'}, \psi_{F_H})_{\mathcal{H}} = F'(\psi_{F_H})$. In other words $H(F') = F'(\psi_{F_H})$, which is nothing but $\mathcal{J}(\psi_{F_H})[F']$. This entails that the map $\mathcal{J}$ is an isomorphism which realizes the isometry between $\mathcal{H}$ and $\mathcal{H}^{**}$, hence $\mathcal{H}$ is reflexive. $\square$

2 Bounded Operators on Hilbert Spaces

In this section we start the study of a notable class of maps between Hilbert spaces whose application to physics are manifold. We shall introduce them gradually, highlighting only the minimal structures which are necessary. Unless stated otherwise we shall work over the field of complex numbers, but all results can be adapted to the real numbers mutatis mutandis.

Definition 19. Let V, V' two complex vector spaces. We call a map $T : V \rightarrow V'$ a **(linear) operator** from V to V' if

$$T(\alpha v + \beta v') = \alpha T(v) + \beta T(v'). \quad \forall v, v' \in V, \text{ and } \forall \alpha, \beta \in \mathbb{C}$$

We call $\mathcal{L}(V, V')$ the collection of all these maps.

The definition can be refined if we assume additionally that we are dealing with Banach spaces:

Definition 20. Let X, Y be two Banach spaces. Then

- We call $\mathcal{B}(X, Y) \subset \mathcal{L}(X, Y)$ the set of all continuous linear operator from X to Y where continuity is defined as per Definition 4. If $Y = X$ then we write $\mathcal{B}(X) \equiv \mathcal{B}(X, X)$ and $\mathcal{L}(X) \equiv \mathcal{L}(X, X)$.
- If $Y = \mathbb{C}$ (or $\mathbb{R}$ if we were working over real vector spaces), we would call $X^* = \mathcal{L}(X, \mathbb{C})$ the **algebraic dual** of X and $X' = \mathcal{B}(X, \mathbb{C})$ the **topological dual** of X .

It is straightforward to see that all these spaces are linear themselves inheriting the vector space structures from X and Y . Nonetheless, the class of all linear operators is certainly huge and it encompasses manifold examples with rather different behaviours. Consider the following simple example:

Example 21. Let $X = Y = l^2(\mathbb{C})$ and let $T : l^2(\mathbb{C}) \rightarrow l^2(\mathbb{C})$ be defined via its action on a generic sequence $z = \{z_n\}_{n \in \mathbb{N}} \in l^2(\mathbb{C})$ as the shift operator

$$T(z_1, z_2, z_3, \dots) = (0, z_1, z_2, \dots).$$

In other words $T(\{z_n\}_{n \in \mathbb{N}}) = \{z'_n\}_{n \in \mathbb{N}}$ where $z'_1 = 0$ while $z'_n = z_{n-1}$ for $n > 1$. If we recall that $\|z\|_{l^2}^2 = \sum_{n=1}^{\infty} |z_n|^2$, a direct computation shows that $\|T(z)\|_{l^2}^2 = \|z\|_{l^2}^2$. Hence, in view of Definition 11, T is an isometry.

As we see T preserves the norm of the Hilbert space element and it is desirable to characterize all the operators which have a similar behaviour.

Theorem 22. Let X, Y be two normed spaces over the complex numbers with norm $\|\cdot\|_X$ and $\|\cdot\|_Y$ respectively. Let $T \in \mathcal{L}(X, Y)$. The following two conditions are equivalent:

1. there exists $K \in \mathbb{R}$ such that $\|T(v)\|_Y \leq K\|v\|_X$ for all $v \in X$,
2. $\sup_{v \in X \setminus \{0\}} \frac{\|T(v)\|_Y}{\|v\|_X} < \infty$

Proof. Suppose 1. holds. Then, for all $v \in V$, $\frac{\|T(v)\|_Y}{\|v\|_X} \leq K < \infty$. Taking the supremum of the left hand side yields the desired result. Suppose now that 2. holds. We can identify K with $\sup_{v \in X \setminus \{0\}} \frac{\|T(v)\|_Y}{\|v\|_X}$ and, per definition of supremum, 1. descends. $\square$

From now on we shall **omit** the subscripts in the norms indicating the space on which they are defined unless there is a risk of confusion. In this case we will reinstate them. The above theorem justifies the following definition:

Definition 23. Let X, Y be normed spaces over the complex numbers. We say that $T \in \mathcal{L}(X, Y)$ is **bounded** if one of the conditions of Theorem 22 are met. We call **operator norm** of T

$$\|T\| = \sup_{v \neq 0} \frac{\|T(v)\|}{\|v\|}.$$

Using the linearity of T is often convenient to rewrite the definition of operator norm in this fully equivalent way:

$$\|T\| = \sup_{v \mid \|v\|=1} \|T(v)\|. \quad (7)$$

Observe that a direct consequence of this definition is that for every bounded operator T between two normed spaces X, Y , it holds that, for every $v \in X$,

$$\|T(v)\| \leq \|T\| \|v\|.$$

It is clear that bounded operators are a nice class of maps between normed spaces, but, as we have seen in Definition 4, we can also require an operator to be continuous, a property which is often desirable as one knows for example from classical mechanics. Luckily enough it turns out that boundedness and continuity of an operator between two normed spaces are two sides of the same coin:

Theorem 24. *Let $T : X \rightarrow Y$ be an operator between two normed spaces over the complex numbers. The following statements are equivalent:*

1. T is continuous at $0 \in X$,
2. T is continuous,
3. T is bounded.

Proof. $3 \implies 1$. This is the simplest implication since, if T is bounded, per linearity, it entails that there exists $K \in \mathbb{R}$ such that $\|T(v - v')\| \leq K\|v - v'\|$, for all $v, v' \in X$. Following Definition 4, for every $\epsilon > 0$, it suffices to take $\delta = \frac{\epsilon}{K}$.

$1 \implies 2$. Suppose that T is continuous at $0 \in X$ and consider $\lim_{v \rightarrow v'} T(v)$. It holds that since the limit is taken with respect to v ,

$$\lim_{v \rightarrow v'} T(v) - T(v') = \lim_{v \rightarrow v'} T(v - v') = \lim_{w \rightarrow 0} T(w) = 0,$$

where $w \doteq v - v'$. Notice that, beside the assumption, we used that T is linear. The second equality can be seen writing the limits in terms of norms.

$2. \implies 3$. Suppose that T is continuous. This is true in particular at $0 \in X$. Hence we can find $\delta > 0$ such that $\|v\| < \delta$ and $\|T(v)\| < 1$ with $v \in X$. Let δ' be such that $0 < \delta' < \delta$ and let $w \in X$. We define $v' \doteq \delta' \frac{w}{\|w\|}$ and we observe that $\|v'\| = \delta' < \delta$. This entails that $\|T(v')\| = \delta' \frac{\|T(w)\|}{\|w\|} < 1$ which implies $\|T(w)\| < \frac{\|w\|}{\delta'}$. This is nothing but condition 1. in Theorem 22 with $K = \frac{1}{\delta'}$. Hence T is bounded. $\square$

To conclude this small excursus, we explain without proving it that there is a reason why the name operator norm is used - see [5, Th. 2.41]:

Proposition 25. *Let X, Y be two normed spaces over the complex numbers. Then the following statements hold true:*

- The operator norm defines a norm over $\mathcal{B}(X, Y)$,
- If Y is a Banach space, then $\mathcal{B}(X, Y)$ is a Banach space itself.
- if Z is a third normed space over the complex numbers than it holds

$$\|\mathbb{I}\| = 1, \quad \|ST\| \leq \|S\|\|T\|, \quad \forall S \in \mathcal{B}(Y, Z) \text{ and } \forall T \in \mathcal{B}(X, Y),$$

where $\mathbb{I}$ is the identity operator on any normed space⁶.

⁶Observe that, if $X = Y = Z$ Proposition 25 entails that $\mathcal{B}(X) \equiv \mathcal{B}(X; X)$ possesses the structure of an algebra.

An additional important consequence which arises from the hypothesis that the target space is a Banach space is the following:

Proposition 26 (Extension of bounded operators). *Let X a normed space over the complex numbers and let Y be a complex Banach space. Let $W \subset X$ be dense and let $T \in \mathcal{B}(W, Y)$, then there exists a unique $\tilde{T} \in \mathcal{B}(X, Y)$ such that $\tilde{T}|_W = T$ and $\|\tilde{T}\| = \|T\|$.*

Proof. Let us show existence. For any $x \in X$, being W dense in X , there exists a sequence $x_n \in W$, $n \in \mathbb{N}$ such that $\lim_{n \rightarrow \infty} x_n = x$. Since T is bounded, it holds that there exists $K > 0$ finite

$$\|T(x_n) - T(x_m)\| \leq K\|x_n - x_m\|.$$

This inequality shows that since $\{x_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence, so it must be $\{T(x_n)\}_{n \in \mathbb{N}}$. Being Y a Banach space per hypothesis, its completeness entails that there exists $\lim_{n \rightarrow \infty} T(x_n)$ which we call $\tilde{T}x$. It is immediate to show that, since $\|\cdot\|$ is a continuous map from Y to $\mathbb{R}$, then the limit does not depend on the approximating sequence. Furthermore, per construction (choose the constant sequence), it holds that $\tilde{T}|_W = T$ and that $\tilde{T}$ is linear. Since $\|T(x_n)\| \leq K\|x_n\|$, by taking the limit, one obtains that $\|\tilde{T}\| \leq K$, hence $\tilde{T} \in \mathcal{B}(X, Y)$.

Let us prove uniqueness. Suppose there exists a second extension U with the same properties of $\tilde{T}$. Take any sequence $\{x_n\}_{n \in \mathbb{N}}$ of elements of W converging to $x \in X$. The restriction constraint entails that $\tilde{T}(x_n) = T(x_n) = U(x_n)$. Hence the sequence $(\tilde{T} - U)(x_n)$ is identically 0 for every finite n . It converges to $0 \in Y$, which entails $\tilde{T}(x) = U(x)$ for all $x \in X$.

To conclude we need to control the norm of $\tilde{T}$. Let $\{x_n\}_{n \in \mathbb{N}}$ be a sequence of elements in W converging to x . It holds

$$\|\tilde{T}(x)\| = \lim_{n \rightarrow \infty} \|T(x_n)\| \leq \|T\| \lim_{n \rightarrow \infty} \|x_n\| = \|T\| \|x\|,$$

from which it descends $\|\tilde{T}\| \leq \|T\|$. At the same time

$$\|\tilde{T}\| = \sup_{x \in X, x \neq 0} \frac{\|\tilde{T}(x)\|}{\|x\|} \geq \sup_{x \in W, x \neq 0} \frac{\|\tilde{T}(x)\|}{\|x\|} = \|T\|,$$

where in the last equality we used that $\tilde{T}|_W = T$ and where we used the definition of operator norm as in item 2. of Theorem 22. $\square$

Although it turns out that most of the physically interesting instances of operators, *e.g.*, the Hamiltonian, are not bounded, there are a few examples which are quite important and enlightening:

1. Every unitary operator U on a Hilbert space $\mathcal{H}$ is bounded. As a matter of fact, since U preserves the norm, for every $\psi \in \mathcal{H}$, it holds $\|U(\psi)\| = \|\psi\|$. By taking the supremum it holds that $\|U\| = 1$. The relevance of this observation stems from the consideration that the time evolution of a dynamical system encoded in the Schrödinger equation or

the action of a continuous symmetry (rotations, translations and so on and so forth) on a Hilbert space are all described via a unitary operator. Although at this stage, it might sound even confusing, bear in mind, that this statement relies strongly on Proposition 26 since the time evolution as well as the action of continuous isometries are almost always defined on dense subspaces of $\mathcal{H}$, rather than on the whole Hilbert space.

2. Consider $\mathcal{H} = L^2(0, 1)$ as well as the so-called **multiplication operator** $\hat{x} : L^2(0, 1) \rightarrow L^2(0, 1)$ such that $\hat{x}(\psi) = x\psi$ for all $\psi \in L^2(0, 1)$. The following inequality holds

$$\|\hat{x}(\psi)\|^2 = \int_0^1 dx |x\psi|^2 \leq \int_0^1 dx |\psi|^2 = \|\psi\|^2.$$

As a consequence, not only $\hat{x}$ is well defined map from $L^2(0, 1)$ into itself, but it is also a bounded operator. By taking the supremum over all unit-norm elements of the Hilbert space, it descends $\|\hat{x}\| \leq 1$. Observe that a straightforward generalization yields that the multiplication operator on $L^2(a, b)$ is bounded whenever $-\infty < a < b < \infty$. In addition $\|\hat{x}\| \leq b - a$. It is also immediate to realize that none of these conclusions, including well-posedness, holds true on $L^2(\mathbb{R})$.

3. Consider $\mathcal{H} = L^2(\mathbb{R}^3)$ as well as the standard Cartesian coordinates on $\mathbb{R}^3$. The **parity transformation** is the map $P : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that $P(x, y, z) = (-x, -y, -z)$. It induces a counterpart on $\mathcal{H}$, namely the **parity operator** $\hat{P} : L^2(\mathbb{R}^3) \rightarrow L^2(\mathbb{R}^3)$ such that $\hat{P}\psi = \psi(P(x, y, z)) = \psi(-x, -y, -z)$ where $\psi(x, y, z)$ stands for any representative of the equivalence class $\psi \in L^2(\mathbb{R}^3)$. It holds

$$\|\hat{P}(\psi)\|^2 = \int_{\mathbb{R}^3} d^3x |\psi(-x, -y, -z)|^2 = \|\psi\|^2,$$

where the last equality descends after the obvious change of coordinates in the integral. As a by-product $\hat{P}$ is a unitary, hence bounded operator. It plays a special role since it is associated in quantum mechanics and, more generally in quantum field theory, to a non continuous symmetry of the underlying quantum system.

Exercise 7. Verify that $\mathcal{B}(X, Y)$ is a normed space whenever endowed with the operator norm.

We conclude quoting some of the most important theorems concerning operators between Banach spaces – see [5, Th. 2.91]:

Theorem 27 (of Banach-Schauder). *Let X, Y be two Banach spaces over the complex numbers and let $T \in \mathcal{B}(X, Y)$ be surjective. Then T is an open map.*

An important corollary of this theorem is the following

Theorem 28 (Banach inverse operator theorem). *Let X, Y be two Banach spaces over the complex numbers and let $T \in \mathcal{B}(X, Y)$ be bijective. Then*

1. $T^{-1} \in \mathcal{B}(Y, X)$,
2. there exists $K' > 0$ such that, for all $v \in X$ $\|T(v)\| \geq K'\|v\|$.

Proof. 1. T^{-1} inherits linearity from that of T , while the Banach-Schauder theorem guarantees us that T is an open map and thus T^{-1} must be continuous since the pre-image of any open set is itself open. Theorem 24 entails that T^{-1} is bounded.

2. Since T^{-1} is bounded, there exists $K > 0$ such that, for all $w \in Y$, $\|T^{-1}(w)\| \leq K\|w\|$. Notice that we have excluded $K = 0$ since it would be in contradiction to the hypothesis of T being invertible. Since T is a bijection, for all $w \in Y$, there exists $v \in X$ such that $w = T(v)$. Hence $\|v\| = \|T^{-1}(T(v))\| \leq K\|T(v)\|$, from which descends the sought conclusion by fixing $K' = K^{-1}$. $\square$

2.0.1 The Closed Graph Theorem

We focus now on a construction related to normed spaces, which will play a role especially in the section devoted to unbounded operators. Let thus X, Y be normed spaces whose norms are $\|\cdot\|_X$ and $\|\cdot\|_Y$. Since these are vector spaces, we can construct their direct sum $X \oplus Y$ and we can endow it naturally with a norm $\|\cdot\|$ defined as follows:

$$\|(x, y)\| \doteq \max\{\|x\|_X, \|y\|_Y\}. \quad \forall (x, y) \in X \oplus Y \quad (8)$$

Exercise 8. Show that the topology induced on $X \oplus Y$ by $\|\cdot\|$ is equivalent to the product topology induced by $\|\cdot\|_X$ and $\|\cdot\|_Y$.

It is also immediate to infer, that, if $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ are Banach spaces, so is $(X \oplus Y, \|\cdot\|)$.

Definition 29. Let X, Y be normed spaces on the complex numbers. Let $T \in \mathcal{L}(X, Y)$ and let

$$G(T) \doteq \{(x, T(x)) \in X \oplus Y \mid x \in X\},$$

be the **graph** of T . We call T **closed** if $G(T)$ is closed in the product topology.

Exercise 9. Prove that $G(T)$ is closed if and only if for any sequence $\{x_n\}_{n \in \mathbb{N}}$ of elements in X such that $\lim_{n \rightarrow \infty} x_n = x \in X$ and such that the sequence $\{T(x_n)\}_{n \in \mathbb{N}}$ converges in Y , we have $\lim_{n \rightarrow \infty} T(x_n) = T(x)$.

We can now state the main theorem of this subsection:

Theorem 30 (Closed graph theorem). Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be Banach spaces over the complex numbers. Then $T \in \mathcal{L}(X, Y)$ is bounded if and only if it is closed.

Proof. Suppose $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ are Banach space and consider $T \in \mathcal{B}(X, Y)$. Then, if $\{x_n\}_{n \in \mathbb{N}}$ is a sequence of elements in X converging to $x \in X$, then $\|T(x_n)\| \leq \|T\| \|x_n\|$, which implies that $\lim_{n \rightarrow \infty} T(x_n) = T(x)$. This entail that $G(T)$ is closed in the product topology, hence T is a closed operator.

Suppose now that T is closed and consider the bilinear map $\pi_1 : G(T) \rightarrow X$ which projects on the first entry, *i.e.*, $\pi_1(x, T(x)) = x$. The projection is a continuous map and, per definition, $\|\pi_1(x, T(x))\| = \|x\|_X \leq \|(x, T(x))\|$, where the inequality is a consequence of (8). In other words π_1 is a bounded map and Banach inverse operator theorem 28 guarantees us that $\pi_1^{-1} : X \rightarrow G(T) \subset X \oplus Y$ is bounded. Consider now the projection on the second element, $\pi_2 : X \oplus Y \rightarrow G(T)$ such that $\pi_2(x, T(x)) = T(x)$. Using the same line of reasoning as for π_1 , we can conclude that π_2 is a continuous, hence bounded, map. At last we observe that we have written T as $\pi_2 \circ \pi_1^{-1}$. Since these are the composition of bounded maps, T is bounded itself. $\square$

2.1 The Adjoint Operator

We examine now one of the main consequences of Riesz theorem leading to the definition of the adjunction of an operator. The relevance of this new operation cannot be emphasized enough since it is at the hearth of the formulation of quantum mechanics and of all quantum theories. In this section we focus only on the case of bounded operators, which do not encompass all physically or mathematically interesting scenarios, but it can be easily tamed and hence it is well-suited for a first approach. Notice that, unless otherwise specified *all Hilbert spaces are indicated as* $\mathcal{H} \equiv (\mathcal{H}, (\cdot, \cdot))$. We will leave the scalar product understood in the definition unless a disambiguation becomes necessary. We recall also that, unless stated otherwise, all Hilbert spaces can be non separable. For all practical purposes, all results are valid without any requirement of separability unless in the proof one has to choose a countable basis or one has to rely on a theorem where such choice has been made.

Proposition 31. *Let $(\mathcal{H}_1, (\cdot, \cdot)_1)$ and $(\mathcal{H}_2, (\cdot, \cdot)_2)$ be two Hilbert spaces and let $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$. There exists a unique operator $T^* : \mathcal{H}_2 \rightarrow \mathcal{H}_1$ such that*

$$(\phi, T\psi)_2 = (T^*\phi, \psi)_1, \quad \forall \psi \in \mathcal{H}_2, \text{ and } \forall \phi \in \mathcal{H}_1.$$

The operator $T^ \in \mathcal{L}(\mathcal{H}_2, \mathcal{H}_1)$ is called the (Hermitian) adjoint (or sometimes Hermitian conjugate) of T .*

Proof. Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be as per hypothesis and let $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$. Fix $\phi \in \mathcal{H}_2$ and consider the map $f_\phi : \mathcal{H}_1 \rightarrow \mathbb{C}$ such that $f_\phi(\psi) \doteq (\phi, T\psi)_2$. We observe that $f_\phi \in \mathcal{B}(\mathcal{H}_1)$. As a matter of facts f_ϕ inherits linearity in its argument from the bilinearity of the Hermitian inner product on $\mathcal{H}_2$. The Cauchy-Schwarz inequality – see Theorem 10 – entails that

$$|f_\phi(\psi)| = |(\phi, T\psi)| \leq \|\phi\|_2 \|T\psi\|_2 \leq \|\phi\|_2 \|T\| \|\psi\|_1,$$

where we used that T is per assumption bounded. Hence dividing each side of the inequality by $\|\psi\|_1$, one obtains

$$\frac{|f_\phi(\psi)|}{\|\psi\|_1} \leq \|T\| \|\phi\|_2.$$

Since ϕ is arbitrary but fixed, we can take the supremum over all $\psi \in \mathcal{H}_1$ to conclude via 2. in Theorem 22 that f_ϕ is a bounded operator. Theorem 24 entails that f_ϕ is continuous. We can thus apply Riesz's Theorem to infer the existence of an element $\Psi_\phi \in \mathcal{H}_1$ such that

$$(\phi, T\psi)_2 = (\Psi_\phi, \psi)_1 \quad \forall \psi \in \mathcal{H}_1$$

In addition the bilinearity of $(\cdot, \cdot)_2$ ensures that the map $\mathcal{H}_2 \ni \phi \mapsto f_\phi : \mathcal{H}_1 \rightarrow \mathbb{C}$ is linear and, consequently, also the map $\mathcal{H}_2 \ni \phi \mapsto \Psi_\phi \in \mathcal{H}_1$. This justifies introducing the linear operator $T^* : \mathcal{H}_2 \rightarrow \mathcal{H}_1$ defined as

$$T^*(\phi) \doteq \Psi_\phi.$$

It remains to be seen that T^* enjoys the wanted properties. First of all, for every $\psi \in \mathcal{H}_1$, it holds:

$$(T^*(\phi), \psi)_1 = (\Psi_\phi, \psi)_1 = (\phi, T\psi)_2.$$

We are left with the proof that T^* is unique. We proceed by contradiction, assuming that there exists $K \in \mathcal{L}(\mathcal{H}_1, \mathcal{H}_2)$ different from T^* such that, for all $\phi \in \mathcal{H}_2$ and for all $\psi \in \mathcal{H}_1$, $(K(\phi), \psi)_1 = (\phi, T\psi)_2$. This implies also that $(K(\phi), \psi)_1 = (T^*(\phi), \psi)_1$ or, in other words, $((K - T^*)(\phi), \psi)_1 = 0$. Since ψ is arbitrary we can choose $\psi = (K - T^*)(\phi)$, hence obtaining

$$((K - T^*)(\phi), (K - T^*)(\phi))_1 = \|(K - T^*)(\phi)\|_1 = 0,$$

where $\|\cdot\|_1$ is the norm induced by $(\cdot, \cdot)_1$. Item 4. in Definition 1 entails that $(K - T^*)(\phi) = 0$. Since ϕ is arbitrary the sought conclusion descends. $\square$

In full analogy with the nomenclature used in the course of linear algebra when discussing maps between finite dimensional vector spaces, given any two vector spaces V, V' and a linear map $T : V \rightarrow V'$, we call

- *Range* of T the set $\text{Ran}(T) \doteq \{v' \in V' \mid \exists v \in V, \text{ for which } v' = T(v)\},$
- *Kernel* of T The set $\ker(T) \doteq \{v \in V \mid T(v) = 0\}.$

The adjoint of a bounded operator between two Hilbert spaces enjoys many additional notable properties besides the already remarkable ones of existence and uniqueness:

Proposition 32. *Let $(\mathcal{H}_1, (\cdot, \cdot)_1)$ and $(\mathcal{H}_2, (\cdot, \cdot)_2)$ be two Hilbert spaces and let $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$. Then:*

1. The **adjunction map** $*$: $\mathcal{B}(\mathcal{H}_1, \mathcal{H}_2) \rightarrow \mathcal{B}(\mathcal{H}_2, \mathcal{H}_1)$ defined as $*(T) = T^*$ for all $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$ is antilinear and involutive, namely $(T^*)^* = T$ for all $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$,
2. The adjoint operator T^* is bounded. Furthermore

$$\|T^*\| = \|T\| \quad \text{and} \quad \|TT^*\| = \|T\|^2 = \|T^*T\|.$$

3. if $\tilde{\mathcal{H}}$ is a third Hilbert space, then, for every $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$ and for every $S \in \mathcal{B}(\tilde{\mathcal{H}}, \mathcal{H}_1)$

$$(TS)^* = S^*T^*.$$

4. $\ker(T) = [\text{Ran}(T^*)]^\perp$ and $\ker(T^*) = [\text{Ran}(T)]^\perp$,

5. T is bijective if and only the same holds true for T^* . In this case $(T^{-1})^* = (T^*)^{-1}$.

Proof. 1. Since an Hermitian inner product is antilinear with respect to first entry, it descends that, for all $\alpha, \beta \in \mathbb{C}$, for all $\psi \in \mathcal{H}_1$, $\phi \in \mathcal{H}_2$ and for all $T, S \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$

$$(\phi, (\alpha T + \beta S)(\psi))_2 = \alpha(T^*(\phi), \psi)_1 + \beta(S^*(\phi), \psi)_2 = ((\bar{\alpha}T^* + \bar{\beta}S^*)(\phi), \psi)_1,$$

which entails $(\alpha T + \beta S)^* = \bar{\alpha}T^* + \bar{\beta}S^*$. In addition the identities

$$(\phi, T(\psi))_2 = (T^*(\phi), \psi)_1 = (\phi, (T^*)^*\psi)_2, \quad \forall \psi \in \mathcal{H}_1, \forall \phi \in \mathcal{H}_2, \text{ and } \forall T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2),$$

combined with the uniqueness of the adjoint as per Proposition 31 entail that $(T^*)^* = T$.

2. For every $\phi \in \mathcal{H}_2$ and for every $\psi \in \mathcal{H}_1$, Theorem 31 together with the Cauchy-Schwarz inequality guarantees that, for every $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$

$$|(T^*(\phi), \psi)_1| = |(\phi, T(\psi))_2| \leq \|\phi\|_2 \|T(\psi)\|_2 \leq \|\phi\|_2 \|T\| \|\psi\|_1,$$

where all norms are those induced from the Hermitian inner products. The arbitrariness of ψ entails that we can fix it as $\psi = T^*(\phi)$. The above inequality becomes

$$\|T^*(\phi)\|_1^2 \leq \|T^*(\phi)\|_1 \|T\| \|\phi\|_2 \implies \|T^*(\phi)\|_2 \leq \|T\| \|\phi\|_2.$$

Dividing each side of the last inequality by $\|\phi\|_2$ and taking the supremum over all $\phi \in \mathcal{H}_2$, we can use 2. in Theorem 22 to conclude that $T^* \in \mathcal{B}(\mathcal{H}_2, \mathcal{H}_1)$ and $\|T^*\| \leq \|T\|$. Having established that T^* is a bounded operator we can reverse the above line of reasoning exchanging the role played by T and T^* . Mutatis mutandis and using, where necessary, that the adjunction map is involutive, one gets to the inequality $\|T\| \leq \|T^*\|$ which, combined with the one already proven, yields $\|T\| = \|T^*\|$.

If we recall that Proposition 25 guarantees has that $\mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$ is a Banach space and that, for all $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$, $\|T^*T\| \leq \|T^*\| \|T\| = \|T\|^2$. At the same time, per Definition 25, the following chain of relations holds true:

$$\begin{aligned} \|T\|^2 &= \left(\sup_{\psi \neq 0} \frac{\|T(\psi)\|_2}{\|\psi\|_1} \right)^2 = \sup_{\psi \neq 0} \frac{(T(\psi), T(\psi))_2}{\|\psi\|_1^2} = \sup_{\psi \neq 0} \frac{(T^*T(\psi), \psi)_1}{\|\psi\|_1^2} \leq \\ &\quad \sup_{\psi \neq 0} \frac{\|T^*T\psi\|_1}{\|\psi\|_1} = \|T^*T\|. \end{aligned}$$

Notice that the inequality is nothing but the application of Theorem 10, while the second equality stems for the realization that the square of the supremum over a set of positive real numbers

coincides with the supremum of the square of the elements of the set. We have thus shown both that $\|T^*T\| \leq \|T\|^2$ and that $\|T\|^2 \leq \|T^*T\|$, hence $\|T\|^2 = \|T^*T\|$. Since $\|T\| = \|T^*\|$, by exchanging the role played by T and T^* and being the adjunction map involutive, one obtains $\|T^*\|^2 = \|TT^*\|$ and thus the sought $\|T\|^2 = \|T^*T\| = \|TT^*\|$.

3. for every $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$, $S \in \mathcal{B}(\tilde{\mathcal{H}}, \mathcal{H}_1)$ and for all $\phi \in \mathcal{H}_2$, $\psi \in \tilde{\mathcal{H}}$, the following chain of equalities holds true:

$$((TS)^*(\phi), \psi)_{\tilde{\mathcal{H}}} = (\phi, TS(\psi))_2 = (T^*(\phi), S(\psi))_1 = (S^*(T^*(\phi)), \psi)_{\tilde{\mathcal{H}}},$$

where $(\cdot, \cdot)_{\tilde{\mathcal{H}}}$ indicates the Hermitian inner product in $\tilde{\mathcal{H}}$. The uniqueness of the adjoint operator, proven in Proposition 31, guarantees that $(TS)^* = S^*T^*$.

4. We prove only the first one since the second follows suit. Suppose that $T \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$ and that $\psi \in \ker(T)$. Then, since $T(\psi) = 0$, for every $\phi \in \mathcal{H}_2$

$$0 = (\phi, T(\psi))_2 = (T^*\phi, \psi)_1.$$

The arbitrariness of ϕ entails that $\psi \in [\text{Ran}(T^*)]^\perp$. Since we have only used identities, we can trace our step back, starting from the assumption that $\psi \in [\text{Ran}(T^*)]^\perp$, proving as a consequence that $\psi \in \ker(T)$.

5. Suppose T is bijective, then Banach inverse operator theorem 28 guarantees that $T^{-1} \in \mathcal{B}(\mathcal{H}_2, \mathcal{H}_1)$. Boundedness implies existence of the adjoint as per Proposition 31. Observing that the adjoint of the identity map $\mathbb{I}$ on a given Hilbert space is the identity itself, the following chain of identities holds true in view of point 3. of this proposition:

$$\mathbb{I} = (\mathbb{I})^* = (TT^{-1})^* = (T^{-1})^*T^* \quad \text{and} \quad \mathbb{I} = (\mathbb{I})^* = (T^{-1}T)^* = T^*(T^{-1})^*.$$

This is tantamount to saying that T^* is invertible, hence a bijection. Furthermore, by direct inspection, it descends that $(T^{-1})^* = (T^*)^{-1}$. The same line of reasoning leads to the sought conclusion starting from the hypothesis that T^* is a bijection. The only difference is the necessity of using also point 1. of this proposition, $(T^*)^* = T$ in particular. $\square$

2.1.1 Further Properties of the Adjoint Operators

We have learned that, given any pair of Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$ and a bounded operator $T : \mathcal{H}_1 \rightarrow \mathcal{H}_2$, there is a canonical way to associate to it an adjoint $T^* \in \mathcal{B}(\mathcal{H}_2, \mathcal{H}_1)$. We can now make a clever use of this information to identify in between the plethora of all possible bounded, linear operators notable subclasses, of paramount relevance especially in quantum mechanics. We recall that, whenever the target and the source Hilbert space coincide, *i.e.*, $\mathcal{H}_1 = \mathcal{H}_2 = \mathcal{H}$, we shall use the notation $\mathcal{B}(\mathcal{H})$ to indicate the collection of bounded, linear operators from $\mathcal{H}$ to itself. From now on we will also omit all subscripts in the norms and in the Hermitian scalar products whenever the space to which they refer will be clear from the context. So... be careful and pay attention!

Definition 33. Let $\mathcal{H}$ and $\mathcal{H}'$ be two Hilbert spaces. Then we call

- **normal** any $T \in \mathcal{B}(\mathcal{H})$ such that $TT^* = T^*T$,
- **self adjoint** any $T \in \mathcal{B}(\mathcal{H})$ such that $T = T^*$
- **isometric** any $T \in \mathcal{L}(\mathcal{H}, \mathcal{H}')$ such that $(T\psi', T\psi)' = (\psi, \psi')$, for all $\psi, \psi' \in \mathcal{H}$
- **unitary** any isometric operator $T \in \mathcal{B}(\mathcal{H}, \mathcal{H}')$, which is moreover surjective.
- **positive** any linear operator $T : \mathcal{H} \rightarrow \mathcal{H}$ such that $(\psi, T\psi) \geq 0$ for all $\psi \in \mathcal{H}$. In this case we say that $T \geq 0$.

Many of these definitions have equivalent formulations, which, depending on the context, might be more useful. We propose two exercises in this direction:

Exercise 10. Let $\mathcal{H}$ and $\mathcal{H}'$ be two Hilbert spaces and let $\mathbb{I}$ be the identity operator on $\mathcal{H}$. Show that $T \in \mathcal{L}(\mathcal{H}, \mathcal{H}')$ is isometric if and only if it is bounded and $T^*T = \mathbb{I}$.

Exercise 11. Let $\mathcal{H}$ and $\mathcal{H}'$ be two Hilbert spaces and let $\mathbb{I}, \mathbb{I}'$ be the respective identity operators. Show that $T \in \mathcal{L}(\mathcal{H}, \mathcal{H}')$ is unitary if and only if it is bounded and $T^*T = \mathbb{I}$, while $TT^* = \mathbb{I}'$.

A useful lemma is the following:

Lemma 34. Let $\mathcal{H}$ be a Hilbert space and let T be a bounded, self-adjoint operator thereon. Then

$$\|T\| = \sup\{ |(\psi, T\psi)|, \mid \psi \in \mathcal{H}, \text{ such that } \|\psi\| = 1 \}. \quad (9)$$

More generally if $T \in \mathcal{L}(\mathcal{H})$ is such that $(\psi, T\psi) = (T\psi, \psi)$ for all $\psi \in \mathcal{H}$ and the right-hand side of (9) is finite, then T is bounded.

Proof. Let $K \doteq \sup\{ |(\psi, T\psi)|, \mid \psi \in \mathcal{H}, \text{ such that } \|\psi\| = 1 \}$. Since we consider unit-norm vectors and since T is bounded, $|(\psi, T\psi)| \leq \|\psi\| \|T\| \|\psi\| = \|T\|$. Hence $K \leq \|T\|$. Since the Hermitian inner product is linear on the second entry and antilinear in the first, it holds that, for all $\psi, \psi' \in \mathcal{H}$

$$4(\psi, T\psi') = (\psi + \psi', T(\psi + \psi')) - (\psi - \psi', T(\psi - \psi')) - i(\psi + i\psi', T(\psi + i\psi')) + i(\psi - i\psi', T(\psi - i\psi')).$$

Taking the real part of both sides⁷ we obtain

$$4\Re(\psi, T\psi') = (\psi + \psi', T(\psi + \psi')) - (\psi - \psi', T(\psi - \psi')).$$

Applying the Cauchy-Schwarz inequality together with the parallelogram rule, we obtain

$$4\Re(\psi, T\psi') \leq K(\|\psi + \psi'\|^2) + K(\|\psi - \psi'\|^2) = 2K(\|\psi\|^2 + \|\psi'\|^2). \quad (10)$$

⁷Observe that, for any Hilbert space $\mathcal{H}$ and for any self-adjoint $T \in \mathcal{L}(\mathcal{H})$ it holds $\overline{(\psi, T\psi)} = (T\psi, \psi) = (\psi, T\psi)$ no matter the choice of $\psi \in \mathcal{H}$.

Here we used that $(\psi + \psi', T(\psi + \psi')) \leq K\|\psi + \psi'\|^2$ because, per definition of K , it holds

$$\left| \left(\frac{\psi + \psi'}{\|\psi + \psi'\|}, T \left(\frac{\psi + \psi'}{\|\psi + \psi'\|} \right) \right) \right| = \left(\frac{\psi + \psi'}{\|\psi + \psi'\|}, T \left(\frac{\psi + \psi'}{\|\psi + \psi'\|} \right) \right) \leq K,$$

and similarly using $\psi - \psi'$. Since both ψ and ψ' in (10) are arbitrary, we can choose ψ' so that $\|\psi'\| = 1$, while $\psi = \frac{T\psi'}{\|T\psi'\|}$ provided that $T\psi' \neq 0$. Since $\|\psi\| = 1$, the inequality (10) becomes

$$4\|T\psi'\| \leq 4K.$$

Notice that, if we would have considered the special case in which $T\psi' = 0$, then this last inequality is automatically satisfied. Using (7), we can take the supremum of the left hand side to conclude that $\|T\| \leq K$.

The last part of the lemma is nothing but the second part of the above proof where it is established that $\|T\| \leq K$, hence that T is bounded. $\square$

Before proceeding with our analysis, three short comments are in due course:

- Given two linear operators $T, U : \mathcal{H} \rightarrow \mathcal{H}$, we can write that $T \geq U$ if $T - U$ is a positive operator,
- All self-adjoint and all unitary operators are normal, but the converse is not true,
- Not all isometric operators from an Hilbert space to itself are unitary as one can readily infer by considering the shift operator defined in Example 21. We leave the details of the computation to the reader.

In view of the first comment we can prove a rather interesting lemma:

Lemma 35. *Let $\mathcal{H}$ be a complex Hilbert space. Then the relation $\geq$ defined out of the notion of positive operators establishes a partial order on $\mathcal{L}(\mathcal{H})$.*

Proof. The relation $\geq$ identified in Definition 33 cannot be of total order since, given two generic $A, B \in \mathcal{L}(\mathcal{H})$, neither $A - B$ nor $B - A$ are necessarily positive. In order to see whether it realizes a partial order, we need to verify the three defining properties of reflexivity, antisymmetry and transitivity.

The first is immediate since, for every $A \in \mathcal{L}(\mathcal{H})$, $A - A = 0$; hence $A \geq 0$. Suppose now that $A, B \in \mathcal{L}(\mathcal{H})$ are such that both $A \geq B$ and $B \geq A$. For every $\psi \in \mathcal{H}$, it holds that both $(\psi, (A - B)\psi)$ and $(\psi, (B - A)\psi)$ are greater or equal to 0. The only option is that $(\psi, (A - B)\psi) = 0$. Using the polarization identity – see Exercise 5 – once replacing ψ with $\psi + \psi'$, $\psi' \in \mathcal{H}$ arbitrary, and once with $\psi + i\psi'$, one establishes that $(\psi', (A - B)\psi) = 0$ for all $\psi, \psi' \in \mathcal{H}$. Choose now a sequence of elements ψ_n such that $\lim_{n \rightarrow \infty} \psi_n = (A - B)\psi$. It holds that $(\psi_n, (A - B)\psi) = 0$. Taking the limit as $n \rightarrow \infty$, thanks to the Cauchy-Schwarz inequality we can conclude that

$$0 = \lim_{n \rightarrow \infty} (\psi_n, (A - B)\psi) = ((A - B)\psi, (A - B)\psi) = \|(A - B)\psi\|^2.$$

In other words $(A - B)\psi = 0$ for all $\psi \in \mathcal{H}$, hence $A = B$.

To conclude we need to establish transitivity. Let $A, B, C \in \mathcal{L}(\mathcal{H})$ be such that $A \geq B$ and $B \geq C$. Then, since the Hermitian inner product is bilinear, it holds that, for all $\psi \in \mathcal{H}$

$$(\psi, (A - C)\psi) = (\psi, (A - B + B - C)\psi) = (\psi, (A - B)\psi) + (\psi, (B - C)\psi) \geq 0.$$

In other words $A \geq C$. □

We stress the hypothesis that $\mathcal{H}$ is a complex Hilbert space, since the above lemma fails if $\mathcal{H}$ is built over the reals. There are several reasons for this fact and finding one is a good exercise:

Exercise 12. Establish why $\geq$ cannot define a partial order on a real Hilbert space.

There is also a notable case where unitary and isometric operators coincide:

Exercise 13. Let $\mathcal{H}$ be a finite dimensional Hilbert space. Show that all isometric operators thereon are also unitary.

We introduce now the simplest, albeit the most important concept of the whole course. Its applications are manifold, being linked especially to the role and to the outcome of a measure in a quantum system. At the same time the definition which follows is nothing but the natural and straightforward generalization of the counterpart learned when studying matrices in linear algebra.

Definition 36. Let V be a complex vector space and let $T \in \mathcal{L}(V)$. We say that $\lambda \in \mathbb{C}$ is an **eigenvalue** of T if there exists an **eigenvector**, namely $v \in V \setminus \{0\}$ such that

$$T(v) = \lambda v.$$

We call **eigenspace** of T relative to the eigenvalue λ , the vector subspace $W_\lambda \subseteq V$, built as the (complex) span of all eigenvectors of λ .

We can now prove a few properties descending from Definition 36:

Proposition 37. Let $\mathcal{H}$ be a Hilbert space and let $T \in \mathcal{B}(\mathcal{H})$ be normal. Then

1. For all $\psi \in \mathcal{H}$ it holds $\|T\psi\| = \|T^*\psi\|$, hence $\ker(T) = \ker(T^*)$ and $\overline{\text{Ran}(T)} = \overline{\text{Ran}(T^*)}$, where the bar stands for the closure to a Hilbert space.
2. λ is an eigenvalue of T with eigenvector ψ if $\bar{\lambda}$ is an eigenvalue of T^* with the same eigenvector,
3. Let λ, λ' be two distinct eigenvalues. Then $W_\lambda \perp W_{\lambda'}$,

Proof. 1. Let T be a bounded, normal operator on $\mathcal{H}$. Hence, for all $\psi \in \mathcal{H}$

$$\|T\psi\|^2 = (T\psi, T\psi) = (T^*T\psi, \psi) = (TT^*\psi, \psi) = (T^*\psi, T^*\psi) = \|T^*\psi\|^2.$$

From this chain of identities, it is immediate to see that if $\psi \in \ker(T)$ then it lies also in $\ker(T^*)$ and vice-versa. Item 4. in Proposition 32 entails that $[\text{Ran}(T^*)]^\perp = [\text{Ran}(T)]^\perp$. To conclude it suffices to recall the generic property of Hilbert spaces, according to which, if we consider any linear subspace W of an Hilbert space $\mathcal{H}$ then $(W^\perp)^\perp = \overline{W}$. In other words $\overline{\text{Ran}(T^*)} = \overline{\text{Ran}(T)}$.

2. Let $\lambda \in \mathbb{C}$. In view of the antilinearity proven in Proposition 32, $(T - \lambda\mathbb{I})^* = T^* - \bar{\lambda}\mathbb{I}$. Furthermore, per direct inspection, $(T - \lambda\mathbb{I})$ is normal since also T is. By item 1. of this proposition, we can infer that, for all $\psi \in \mathcal{H}$

$$\|(T - \lambda\mathbb{I})\psi\| = \|(T^* - \bar{\lambda}\mathbb{I})\psi\|,$$

from which it descends the sought statement since, if $(T - \lambda\mathbb{I})\psi = 0$, then also $(T^* - \bar{\lambda}\mathbb{I})\psi$ vanishes.

3. Let $\lambda, \lambda' \in \mathbb{C}$ be eigenvalues relative respectively to the eigenvectors $\psi, \psi' \in \mathcal{H}$. Using item 2. of this proposition, it holds:

$$\lambda(\psi', \psi) = (\psi', T\psi) = (T^*\psi', \psi) = (\bar{\lambda}'\psi', \psi) = \lambda'(\psi', \psi).$$

Since, per hypothesis $\lambda \neq \lambda'$, the only possibility is that $(\psi', \psi) = 0$, from which the sought result descends. $\square$

Corollary 38. *Let $\mathcal{H}$ be a Hilbert space and let $T \in \mathcal{L}(\mathcal{H})$. Then*

1. *If $T \geq 0$, then its eigenvalues are real and non-negative,*
2. *If T is bounded and self-adjoint, then its eigenvalues are real,*
3. *if T is isometric its eigenvalues λ are such that $|\lambda| = 1$.*

Proof. 1. Suppose that T is positive and that $\lambda \in \mathbb{C}$ is an eigenvalue relative to the eigenvector $\psi \in \mathcal{H}$. Then

$$0 \leq (\psi, T\psi) = \lambda(\psi, \psi) = \lambda\|\psi\|^2.$$

This is possible only if λ is real and non-negative.

2. Suppose that T is bounded and self-adjoint and let λ be an eigenvalue relative to the eigenvector $\psi \in \mathcal{H}$. Then, by item 2. of Proposition 37

$$\lambda(\psi, \psi) = (\psi, T\psi) = (T\psi, \psi) = \bar{\lambda}(\psi, \psi).$$

Since $\psi \neq 0$, it holds that $\lambda = \bar{\lambda}$.

3. Suppose that T is isometric and that λ is an eigenvalue relative to the eigenvector $\psi \in \mathcal{H}$. Then

$$|\lambda|^2(\psi, \psi) = (T\psi, T\psi) = (\psi, \psi).$$

Since $\psi \neq 0$, it descends $|\lambda| = 1$. $\square$

This corollary guarantees us that, when working with bounded operators, the self-adjoint and the positive ones have real eigenvalues. In view of the quantum mechanical interpretation of the eigenvalues of an operator as the possible results of a measurement process of a system, it becomes of paramount importance to establish criteria to identify operators whose eigenvalues are real. The next proposition is a first step in this direction, with the net advantage that it tells us that positive bounded operators are just a special class of the self-adjoint ones:

Proposition 39. *Let $\mathcal{H}$ be a Hilbert space. Then the following statements hold true:*

1. *If $T \in \mathcal{L}(\mathcal{H})$ is such that $(\psi', T\psi) = (T\psi', \psi)$ for all $\psi, \psi' \in \mathcal{H}$, then T is bounded and self-adjoint,*
2. *if $T \in \mathcal{B}(\mathcal{H})$ is such that $(\psi, T\psi) = (T\psi, \psi)$ for all $\psi \in \mathcal{H}$, then T is self-adjoint,*
3. *if $T \in \mathcal{B}(\mathcal{H})$ is positive, then it is also self-adjoint.*

Proof. 1. It suffices to show that T is bounded, since the uniqueness of the adjoint proven in Proposition 31 entails that, in this case, $T = T^*$. Boundedness is a by-product of the *closed graph theorem* 30. We focus only on the remaining items.

2. The hypothesis entails that, for any $\psi \in \mathcal{H}$, $((T - T^*)\psi, \psi) = 0$. The same argument used in Lemma 35 guarantees that this entails $\|(T - T^*)\psi\| = 0$, hence $T = T^*$.

3. Suppose now that T is positive. Then, for any $\psi \in \mathcal{H}$ it holds that $(\psi, T\psi) \geq 0$ and in particular it is real. Thus it coincides with its complex conjugate. By the third property in Definition 9, this is tantamount to saying that

$$(\psi, T\psi) = (T\psi, \psi).$$

Hence we are back to the case analysed in item 2. □

2.1.2 Orthogonal Projectors

We need one last elementary notion related to boundary operators before dwelling into more technical results:

Definition 40. *Let $\mathcal{H}$ be a Hilbert space. We say that $P \in \mathcal{B}(\mathcal{H})$ is an **orthogonal projector** if $P^2 = P$ and $P^* = P$.*

Lemma 41. *Let $\mathcal{H}$ be a Hilbert space and let P be an orthogonal projector. Then P is a positive operator.*

Proof. Let $\psi \in \mathcal{H}$ be arbitrary. Then it holds

$$(\psi, P\psi) = (\psi, P^2\psi) = (P^*\psi, P\psi) = (P\psi, P\psi) = \|P\psi\|^2 \geq 0,$$

which entails that $P \geq 0$ since P is bounded per construction. □

One of the nice properties of orthogonal projectors is the possibility of characterizing them with great detail. Since they are of extreme usefulness in analysing self-adjoint operators, this is a welcome feature:

Theorem 42. *Let $\mathcal{H}$ be a Hilbert space and let $P \in \mathcal{B}(\mathcal{H})$ be an orthogonal projector. Let $W \doteq P[\mathcal{H}]$. The following holds:*

1. $Q = \mathbb{I} - P$ is an orthogonal projector, being $\mathbb{I}$ the identity on $\mathcal{H}$,
2. $Q(\mathcal{H}) = W^\perp$ and $\mathcal{H} = W \oplus W^\perp$,
3. For all $\psi \in \mathcal{H}$, $\|\psi - P(\psi)\| = \min\{\|\psi - \psi'\|; \psi' \in W\}$,
4. Let e_i , $i \in I$ be a basis of W with the index i running over a set I . Then

$$P(\cdot) = s - \sum_{i \in I} e_i(e_i, \cdot),$$

where s — entails that the convergence of the sum is meant with respect to the strong topology.

5. $\mathbb{I} \geq P$ and, unless $P = 0$, then $\|P\| = 1$.

Proof. 1. We observe that $Q^2 = (\mathbb{I} - P)^2 = \mathbb{I} + P^2 - 2P = \mathbb{I} - P$ since P is a projector per hypothesis. Furthermore $Q^* = (\mathbb{I} - P)^* = \mathbb{I}^* - P^* = Q$. Hence Q is an orthogonal projector

2. Suppose $\psi' = Q(\tilde{\psi})$, $\tilde{\psi} \in \mathcal{H}$. Then, for every $\psi \in W$ it holds

$$(\psi, \psi') = (\psi, Q\tilde{\psi}) = (Q^*\psi, \tilde{\psi}) = (\psi - P\psi, \tilde{\psi}) = 0,$$

where we used that $Q^* = Q$ and that $P\psi = \psi$ since $\psi \in W$. We have shown that $Q(\mathcal{H}) \subseteq W^\perp$. Suppose now that W is a closed subspace and that $\mathcal{H} = W \oplus Q(\mathcal{H})$. Since the theory of Hilbert spaces guarantees also that $\mathcal{H} = W \oplus W^\perp$, it descends that $Q(\mathcal{H}) = W^\perp$ per the uniqueness of the decomposition. We show first that W is closed. Let $\{\psi_n\}_{n \in \mathbb{N}}$ be such that $P(\psi_n)$ is a Cauchy sequence. Since Hilbert spaces are complete, there must exist $\Psi \in \mathcal{H}$ such that $\lim_{n \rightarrow \infty} P(\psi_n) = \Psi$. Since P is bounded, hence continuous per Proposition 24, we can apply P to the last identity, exchanging it with the limits to obtain $\lim_{n \rightarrow \infty} P(\psi_n) = P(\Psi)$ since $P^2 = P$. Hence $\Psi = P\Psi$ and $\Psi \in W$. We have shown that W is complete, thus a closed subspace. Let $\psi \in \mathcal{H}$ be arbitrary; we can write $\psi = w + v$, where $w = P(\psi) \in W$ and $v = Q(\psi) \in Q(\mathcal{H})$. We need to show that this decomposition is unique which is tantamount to proving that $W \cap Q(\mathcal{H}) = \emptyset$.

Suppose there exists $\phi \in W \cap Q(\mathcal{H})$. Then $\phi = P\phi = Q\phi$ which entails $2P\phi = \phi$. Since $\phi = P\phi$, it descends $P\phi = 0$, hence $\phi = 0$.

3. and 4. These are technical results which we omit. The reader can refer to [5].

5. Let $\psi \in \mathcal{H}$ be arbitrary and let P be an orthogonal project. Then

$$0 \leq (Q\psi, Q\psi) = (\psi, Q^2\psi) = (\psi, Q\psi) = \|\psi\|^2 - (\psi, P\psi).$$

Hence $\mathbb{I} \geq P$ and $(P\psi, P\psi) = (\psi, P\psi) \leq \|\psi\|^2$. By (7), this entails taking the supremum over all unit norm $\psi \in \mathcal{H}$ that $\|P\| \leq 1$. Yet, if $P \neq 0$, there must exist $\psi' \in W$ and without loss of generality we can take it of unit norm. Hence, since $P\psi' = \psi'$, we have found a vector for which $\|P\psi\| = 1$, hence $\|P\| = 1$. $\square$

The next statement shows that the requirement of orthogonality for projectors does not entail any loss of generality:

Proposition 43. *Let $\mathcal{H}$ be a Hilbert space and let W be a closed subspace. Let $P : \mathcal{H} \rightarrow W$ and $Q : \mathcal{H} \rightarrow W^\perp$ be the associated projectors. They are orthogonal.*

Proof. Since W is closed subspace, $\mathcal{H} = W \oplus W^\perp$. Hence $Q = \mathbb{I} - P$ where $\mathbb{I}$ is the identity operator on $\mathcal{H}$. It suffices thus to prove that $P^* = P$. Let $\psi, \psi' \in \mathcal{H}$ be arbitrary and let $w, w' \in W$ and $v, v' \in W^\perp$ be the unique elements for which $\psi^{(t)} = w^{(t)} + v^{(t)}$. Then

$$(\psi', P\psi) = (w' + v', w) = (w', w).$$

At the same time

$$(P\psi', \psi) = (w', w + v) = (w', w),$$

from which it descends that $(P\psi', \psi) = (\psi', P\psi) = (P^*\psi', \psi)$ where we used the definition of adjoint in the last identity. It descends $((P - P^*)\psi', \psi) = 0$ for all $\psi \in \mathcal{H}$, in particular if $\psi = (P - P^*)\psi'$. This entails $\|(P - P^*)\psi'\|^2 = 0$ for all $\psi' \in \mathcal{H}$. Hence $P\psi' = P^*\psi'$, i.e., $P = P^*$. $\square$

2.2 The Square Root of a Positive Operator

Positive, bounded operators enjoy several special properties, the most notable being that they are always self-adjoint and their eigenvalues lie in $[0, \infty)$. Additionally we can perform on them certain operations which are in general ill-defined. We review here the most important one:

Definition 44. *Let $\mathcal{H}$ be a Hilbert space and let $A \in \mathcal{B}(\mathcal{H})$. We say that $B \in \mathcal{B}(\mathcal{H})$ is a **square root** of A if $A = B^2$. Furthermore if $B \geq 0$ we call it a **positive root**.*

Positive roots are often indicated also as $\sqrt{A}$, but it is clear that, in general, roots might not be well-defined. We shall give sufficient conditions for their existence. As a preliminary result we need an ancillary proposition, whose proof we omit, being technical even for the taste of the author [5, Prop. 3.65]:

Proposition 45. *Let $\mathcal{H}$ be a Hilbert space and let $\{A_n\}_{n \in \mathbb{N}}$ be a non decreasing (non-increasing) sequence of bounded, self-adjoint operators thereon. If $\{A_n\}_{n \in \mathbb{N}}$ is bounded from below (above) by $K \in \mathcal{B}(\mathcal{H})$, then there exists a bounded, self-adjoint operator A on $\mathcal{H}$ such that $A \leq K$ ($A \geq K$) and*

$$s - \lim_{n \rightarrow \infty} A_n = A.$$

We can now state the main result of this section:

Theorem 46. *Let $\mathcal{H}$ be an Hilbert space and let $A \in \mathcal{B}(\mathcal{H})$ be positive. Then there exists a unique positive square root $\sqrt{A}$ such that*

- (a) *it commutes with any bounded operator B on $\mathcal{H}$, which, in turn, commutes with A ,*
- (b) *it is bijective is also A is bijective.*

Proof. Just kidding... have a look at [5, Th. 3.66] and you will see what I mean! $\square$

Definition 47. *Let $\mathcal{H}$ be a Hilbert space and let $A \in \mathcal{B}(\mathcal{H})$. We call **modulus** of A the bounded, positive and self adjoint operator*

$$|A| = \sqrt{A^*A}.$$

Corollary 48. *Let $\mathcal{H}$ be a Hilbert space and let $A \in \mathcal{B}(\mathcal{H})$. Then its modulus enjoys the following properties:*

- a) $\ker(|A|) = \ker(A)$,
- b) $\overline{\text{Ran}(|A|)} = (\ker(A))^\perp$

Proof. We start from a). For any $\psi \in \mathcal{H}$ it holds that

$$\| |A|\psi \|^2 = (|A|\psi, |A|\psi) = (\psi, |A|^2\psi) = (\psi, A^*A\psi) = (A\psi, A\psi) = \|A\psi\|^2, \quad (11)$$

where we used that $|A|$ is self-adjoint. From the identities, it is manifest that $|A|\psi = 0$ if and only if $A\psi = 0$.

Let us focus on b). In view of the theory of Hilbert spaces we know that, for any subspace $W \in \mathcal{H}$, $\overline{W} = (W^\perp)^\perp$. Therefore $\overline{\text{Ran}(|A|)} = ((\text{Ran}(|A|))^\perp)^\perp$. At the same time $(\text{Ran}(|A|))^\perp = \ker(|A|)$. As a matter of facts, suppose $\psi \in (\text{Ran}(|A|))^\perp$; for any $\psi' \in \mathcal{H}$

$$0 = (\psi, |A|\psi') = (|A|\psi, \psi').$$

Since ψ' is arbitrary, the only option is $\psi \in \ker(|A|)$. Reading this equality from right to the left, we realize that also the converse holds true, namely if $\psi \in \ker(|A|)$, then $\psi \in (\text{Ran}(|A|))^\perp$. Putting together these results, we obtain $\overline{\text{Ran}(|A|)} = (\ker(|A|))^\perp$. Using item a) of this corollary we obtain the sought conclusion. $\square$

Theorem 49 (Polar Decomposition). *Let $\mathcal{H}$ be a Hilbert space and let $A \in \mathcal{B}(\mathcal{H})$. Then there exists a unique pair of operators $P, U \in \mathcal{B}(\mathcal{H})$ such that*

- a) $A = UP$,
- b) $P \geq 0$,
- c) U is isometric on $\text{Ran}(P)$,
- d) $U|_{\ker(P)} = 0$.

Furthermore it holds that

- 1. $P = |A|$ and thus $\ker(U) = \ker(A) = \ker(P) = [\text{Ran}(P)]^\perp$,
- 2. if A is bijective, then $U = A|A|^{-1}$.

The splitting $A = UP$ is called the **polar decomposition** of A .

Proof. We start proving uniqueness of the decomposition. Let us focus first of all on P . If any decomposition $A = UP$ with the given properties exists, then $A^* = (UP)^* = PU^*$ since P is positive and bounded, hence self-adjoint per Proposition 39. As a by-product $A^*A = PU^*UP$. Since U is isometric on $\text{Ran}(P)$, it holds that, for all $\psi, \psi' \in \mathcal{H}$

$$(UP\psi, UP\psi') = (P\psi, P\psi') \implies (\psi, PU^*UP\psi') = (\psi, P^2\psi'),$$

which entails $A^*A = P^2$. Hence, for any decomposition as per hypothesis, the positive operator P is the unique positive square root of A^*A , namely $|A|$.

We prove uniqueness of U . As a starting point, since $\mathcal{H} = \ker(P) \oplus (\ker(P))^\perp$ and since P is self-adjoint being positive, we can use item 4. in Proposition 32 to conclude that

$$\mathcal{H} = \ker(P) \oplus [\text{Ran}(P^*)^\perp]^\perp = \ker(P) \oplus \overline{\text{Ran}(P^*)} = \ker(P) \oplus \overline{\text{Ran}(P)}.$$

In view of this decomposition, defining an operator on $\mathcal{H}$ unambiguously is tantamount to specifying its image on $\ker(P)$ and $\overline{\text{Ran}(P)}$. We set $U|_{\ker(P)} = 0$, while $U|_{\text{Ran}(P)}$ is defined via the relation $UP\psi = A\psi$ for every $\psi \in \mathcal{H}$. Since U is bounded per construction, so it is on $\text{Ran}(P)$ and thus it can be extended uniquely to the closure $\overline{\text{Ran}(P)}$ on account of Proposition 26. In other words U is unique. Observe that since $P = |A|$, it holds that, for any $\psi \in \mathcal{H}$, $A\psi = U(|A|\psi)$. If we combine this identity with (11) guaranteeing that $\|A\psi\| = \||A|\psi\|$, we can infer that U is an isometry on $\text{Ran}(P)$. A direct inspection of the above argument entails that, besides uniqueness, we have also shown existence as well as item 1.

We are left with 2. We observe that, if A is injective, item 1. guarantees that also U is injective. At the same time the decomposition $A = UP$ guarantees that $\text{Ran}(A) \subseteq \text{Ran}(U)$ which is tantamount to saying that U is surjective if A is. Hence if A is bijective, U is a bijective isometry, in other words a unitary operator. At last, since $P = |A|$, it holds $A = U|A|$. Since both A and U are bijective, so must be $|A|$. Hence it admits an inverse and it is meaningful to write $U = A|A|^{-1}$.

□

3 Compact Operators

In this section we introduce and we discuss the properties of a notable class of operators. They are of paramount relevance especially on infinite dimensional Hilbert spaces since they represent the closest possible extension of what matrices are on finite dimensional Hilbert spaces. We divide the section in two main parts: The first subsection introduce the main definition and properties, making only use of normed spaces. The reader, who is not familiar with the basic aspects of topology, is strongly advised to revise them. In the second part we will focus mainly on Hilbert spaces and, although, not always necessary, we will make the assumption that all spaces are separable.

3.1 Compact Operators on Normed Spaces

In this section we consider normed spaces which are a special instance of topological spaces with the topology induced by the norm itself, see Appendix A. A preliminary definition is the following:

Definition 50. *Let X be a normed space. A subset $W \subset X$ is said to be **sequentially compact** if and only if every sequence $\{x_n\}_{n \in \mathbb{N}}$ of elements in W has a subsequence converging in W .*

Proposition 51. *Let X be a normed space and let $W \subseteq X$. If any sequence of elements in W admits a converging subsequence, then W is relatively compact. The limit might not lie in W .*

Proof. W is relatively compact if $\overline{W}$ is compact. In view of Definition 50 it suffices to prove that for every sequence $\{x_n\}_{n \in \mathbb{N}}$ of elements in $\overline{W}$, there exists a subsequence converging in $\overline{W}$. Consider any such sequence and fix n . Per definition of closure there must exist a sequence of elements $\{y_k^{(n)}\}$ in W converging to x_n as $k \rightarrow \infty$. Considering a sufficiently large $\bar{k}$, we can construct a third sequence

$$z_n \doteq y_{\bar{k}}^{(n)}, \quad \|x_n - z_n\| < n^{-1}.$$

Per assumption we can extract a subsequence $\{z_{n_p}\}_{n_p \in \mathbb{N}}$ converging to an element $x \in \overline{W}$. Furthermore

$$\|x_{n_p} - x\| \leq \|x_{n_p} - z_{n_p}\| + \|z_{n_p} - x\|,$$

where the left hand side can be made smaller at leisure since $\|x_{n_p} - z_{n_p}\| < n_p^{-1}$. In other words $x_{n_p} \rightarrow x$ as $n_p \rightarrow \infty$. $\square$

A technical corollary whose proof requires Baire's theorem, which we have not discussed (hence we omit the proof), is the following

Corollary 52. *Let X be an infinite dimensional normed space. Then*

- *if $W \subset X$ is compact, then the interior of W is empty,*
- *if X is a Banach space, then X cannot be obtained as a countable union of compact subsets.*

Before moving to the theory of operators, we remark that an important consequence of these results is the proof of one key difference between finite and infinite dimensional normed spaces. As shown in [5, Prop. 4.5], on any, infinite dimensional, normed space the closure of any open ball of any centre and of any radius cannot be compact. On the contrary, in finite dimension, this is the case as one can show by using the Heine-Borel theorem.

Definition 53. *Let X be a normed space. We say that $W \subset X$ is **bounded** if there exist $x \in X$ and $\delta > 0$ such that $W \subset B_\delta(x) \doteq \{x' \in X \mid \|x - x'\| < \delta\}$.*

Definition 54. *Let X, Y be normed spaces over the complex numbers. We say that $T \in \mathcal{L}(X, Y)$ is a **compact operator** if one of these two conditions is met:*

1. For any bounded subset $W \subset X$, $T(W)$ is relatively compact in Y ,
2. If $\{x_n\}_{n \in \mathbb{N}}$ is a bounded sequence of elements in X , there exists a subsequence $\{x_{n_k}\}_{n_k \in \mathbb{N}}$ such that $T(x_{n_k})$ converges in Y .

The set of all compact operators is indicated with $\mathcal{B}_\infty(X, Y)$ or with $\mathcal{B}_\infty(X)$ if $X = Y$.

Notice that, while the first condition automatically implies the second, the converse is a consequence of Proposition 51. We prove a few notable properties of compact operators, both individually and as a set:

Lemma 55. *Let X, Y be normed spaces over the complex numbers. Then any compact operator $T : X \rightarrow Y$ is also bounded, that is $\mathcal{B}_\infty(X, Y) \subset \mathcal{B}(X, Y)$.*

Proof. Consider $B_1(0)$ the unit ball in X centred at 0. Since T is compact, $T(B_1(0))$ is relatively compact. Its closure can be thus covered with a sufficient number, say N , of balls $B_r(y_i)$, $i = 1, \dots, N$. Hence $\overline{T(B_1(0))} \subset \bigcup_{i=1}^N B_r(y_i) \subset B_{R+r}(0)$ where R is the maximum distance between the centres y_i and the origin 0 in Y . As a by-product, for any $v \in X$ such that $\|v\| = 1$, $\|T(v)\| \leq R + r$, hence $\|T\| \leq r + R$. $\square$

Proposition 56. *Let X, Y be normed spaces over the complex numbers and let $\mathcal{B}_\infty(X, Y)$ be the set of compact operators. Then*

- (a) $\mathcal{B}_\infty(X, Y)$ is a vector subspace of $\mathcal{B}(X, Y)$.
- (b) if Z is normed subspace and $A \in \mathcal{B}_\infty(X, Y)$, then⁸ $AB \in \mathcal{B}_\infty(Z, Y)$ for all $B \in \mathcal{B}(Z, X)$ and $CA \in \mathcal{B}_\infty(X, Z)$ for all $C \in \mathcal{B}(Y, Z)$.
- (c) Let Y be a Banach space and let $\{A_n\}_{n \in \mathbb{N}}$ be a sequence of compact operators from X to Y converging uniformly to a bounded operator A . Then $A \in \mathcal{B}_\infty(X, Y)$. In other words $\mathcal{B}_\infty(X, Y)$ is a closed subspace of $\mathcal{B}(X, Y)$ in the norm topology.

The proof of this proposition is very long and we omit it. The interested reader can refer to [5, Prop. 4.9].

Exercise 14. Prove that the identity operator cannot be a compact operator on an infinite dimensional normed space.

Lemma 57 (Banach). *Let x_i with i running over a finite or infinite set of indices be a sequence of linearly independent vectors in a normed space X . Then there exists a corresponding sequence of elements y_j in X such that, for all admissible integers n*

1. $\|y_n\| = 1$ for all n ,
2. $y_n \in X_n \doteq \text{span}_{\mathbb{C}}(x_1, \dots, x_n)$,

⁸In other words we are proving that, for any normed space X , $\mathcal{B}_\infty(X)$ is a (non-unital) algebra.

$$3. d(y_n, X_{n-1}) \doteq \inf_{x \in X_{n-1}} \|x - y_n\| > \frac{1}{2}.$$

Proof. First of all we observe that $d(y_n, X_{n-1})$ exists and it is finite since it is the infimum of a non empty set of real numbers, bounded from below by 0. We build the sought sequence as follows: Let $y_1 \doteq \frac{x_1}{\|x_1\|}$. Since the vectors x_i are linearly independent for all values of i , we know that $x_n \notin X_{n-1}$ and $d(x_n, X_{n-1}) = k > 0$. Choose $x' \in X_{n-1}$ so that $k < \|x_n - x'\| < 2k$. Since $x' \in X_{n-1}$, $k = d(x_n, X_{n-1}) = d(x_n - x', X_{n-1})$. We define $y_n \doteq \frac{x_n - x'}{\|x_n - x'\|}$. Per direct inspection and per construction the new sequence y_n satisfies all three conditions. $\square$

Theorem 58. *Let X be a normed space and let $T \in \mathcal{B}_\infty(X)$. Then*

1. $\forall \delta > 0$, there exists a finite number of eigenspaces of T with eigenvalue λ such that $|\lambda| > \delta$,
2. If $\lambda \neq 0$ is an eigenvalue of T with eigenspace W_λ , then $\dim(W_\lambda) < \infty$,
3. The eigenvalues of T form a bounded, at most countable, set. They can be ordered so that $|\lambda_1| \geq |\lambda_2| \geq \dots \geq 0$, where 0 is the only possible limit point.

Proof. 1. and 2. If the normed space is a finite dimensional set the theorem is basically a trivial consequence of standard results from linear algebra. Hence we assume that X is infinite dimensional. Let $\lambda_1, \lambda_2, \dots$ be a sequence of distinct eigenvalues, possibly with multiplicities, such that, for any arbitrary, but fixed $\delta > 0$, there exists $\bar{i} \in \mathbb{N}$ such that $|\lambda_i| > \delta$ for all $i > \bar{i}$. Suppose there exists a corresponding infinite sequence of linearly independent eigenvectors $x_1, x_2, \dots$. Using Banach Lemma 57, we can construct a corresponding sequence $y_1, y_2, \dots$ enjoying all properties proven in the lemma. Consider the associated sequence $\{\frac{y_n}{\lambda_n}\}_{n \in \mathbb{N}}$. It is bounded since $|\lambda_n| > \delta$ for all n . Since T is a compact operator, there must exist a converging subsequence of $\{T(\frac{y_n}{\lambda_n})\}_{n \in \mathbb{N}}$. Notice that per construction, for each index n in the sequence, there exists $\beta_k \in \mathbb{C}$, $k = 1, 2, \dots, n$ such that $y_n = \sum_{k=1}^n \beta_k x_k$. Since each x_k is an eigenvector of T with eigenvalue λ_k

$$T\left(\frac{y_n}{\lambda_n}\right) = \sum_{k=1}^n \beta_k \frac{\lambda_k}{\lambda_n} x_k = y_n + z_n,$$

where, per direct inspection $z_n = \sum_{k=1}^{n-1} \beta_k \left(\frac{\lambda_k}{\lambda_n} - 1\right) x_k$. Let i, j be two indices in the sequences that we are considering and assume that $i > j$. Then

$$\left\|T\left(\frac{y_i}{\lambda_i}\right) - T\left(\frac{y_j}{\lambda_j}\right)\right\| = \|y_i + z_i - y_j - z_j\| > \frac{1}{2},$$

where, using the notation of Banach lemma, $y_j - z_j - z_i \in X_{i-1}$. Item 3. in Lemma 57 entails the above inequality. We have found a contradiction to the hypothesis that $T \in \mathcal{B}_\infty(X)$, since there cannot exist a converging subsequence of $\{T(\frac{y_n}{\lambda_n})\}_{n \in \mathbb{N}}$. Hence there cannot exist infinite linearly independent eigenvectors with eigenvalues greater than any, but fixed, strictly positive number. This proves both 1. and 2.

3. This is a straightforward consequence of item 1., choosing a sequence $\delta_n = n^{-1}$. For each, but fixed $n \in \mathbb{N}$, we have a finite number of eigenvalues, which thus are a bounded set which can be ordered increasingly with respect to their absolute value. By taking $n \rightarrow \infty$, at most we get a countable set, but still bounded. $\square$

3.2 Compact Operators on Hilbert Spaces and their Spectral Representation

In this section, we specialize our analysis of compact operators to the case when the normed space is actually a Hilbert space. A first alternative characterization of compact operators is the following:

Proposition 59. *Let $\mathcal{H}$ be a Hilbert space and let $T \in \mathcal{B}(\mathcal{H})$. Then T is compact if and only if $|T|$ – see Definition 47 – is compact.*

Proof. Suppose T is compact and let $\{x_n\}_{n \in \mathbb{N}}$ be a bounded sequence in $\mathcal{H}$. Since T is compact, we can extract a converging sequence $\{T(x_{n_k})\}_{n_k \in \mathbb{N}}$. Consider now $|T|(x_{n_k})$; (11) guarantees that $\||T|(x_{n_k})\| = \|T(x_{n_k})\|$. Hence also the sequence $|T|(x_{n_k})$ is converging. Exchanging the role of T and $|T|$ we can repeat the same line of reasoning, hence proving the proposition. $\square$

We recall that all properties proven in the previous part of the section are inherited by compact operators on Hilbert spaces, especially that $\mathcal{B}_\infty(\mathcal{H})$ is a vector subspace of $\mathcal{B}(\mathcal{H})$ and that the product between a compact and a bounded operator is still compact – see Proposition 56. In a more refined language this entails that $\mathcal{B}_\infty(\mathcal{H})$ is a two-sided ideal of the algebra of bounded operators. Hence it is legitimate to consider the quotient between these vector spaces and resulting space inherits the algebra structure. Actually we can prove something even stronger which emphasizes the role of compact operators within the class of bounded ones:

Proposition 60. *Let $\mathcal{H}$ be a Hilbert space and let $T \in \mathcal{B}_\infty(\mathcal{H})$. Then $T^* \in \mathcal{B}_\infty(\mathcal{H})$.*

Proof. Let us consider $T \in \mathcal{B}_\infty(\mathcal{H})$. Proposition 59 entails that also $|T| \in \mathcal{B}_\infty(\mathcal{H})$. Using the polar decomposition, seen in Theorem 49, we know that there exists $U \in \mathcal{B}(\mathcal{H})$ such that $T = U|T|$. Per construction the modulus of a bounded operator is self-adjoint. Hence $T^* = |T|U^*$. Yet item 2. of Proposition 32 guarantees us that U^* is bounded. In other words we have written T^* as the product of a compact and of a bounded operator. Hence, in view of Proposition 56, it is a compact operator. $\square$

We can now prove the first and the simplest instance of one of the most important structural aspects of a large class of operators on Hilbert spaces, namely the so-called *spectral decomposition*, often referred to as *spectral theorem* or *spectral calculus*.

Theorem 61 (Hilbert). *Let $\mathcal{H}$ be an Hilbert space and let T be a self-adjoint, compact operator thereon. Then, every eigenspace of T with non-vanishing eigenvalue has finite dimension. Furthermore, if we call $\sigma_p(T)$ the collection of all eigenvalues, it holds:*

1. $\sigma_p(T)$ is a non empty, real set,

2. $\|T\| = \sup\{|\lambda| \mid \lambda \in \sigma_p(T)\},$
3. $T = 0$ if and only if 0 is the only eigenvalue.

Proof. Let us start by proving that every eigenspace of T with non vanishing eigenvalue $\lambda \in \mathbb{C}$ is finite dimensional. The idea is to use [5, Prop. 4.5], according to which, on any normed space the closure of any open ball of any centre and of any radius cannot be compact if the space has infinite dimension. Let $\mathcal{H}_\lambda$ be the eigenspace relative to λ and let $\mathcal{B}_1(0) \subset \mathcal{H}_\lambda$ be the unit ball centred at the origin in $\mathcal{H}_\lambda$. Notice that $\lambda^{-1}\mathcal{B}_1(0)$ is bounded per construction, hence, being T a compact operator $\mathcal{B}_1(0) = T(\lambda^{-1}\mathcal{B}_1(0))$ is relatively compact. This means that $\overline{\mathcal{B}_1(0)}$ is compact, but, in view of [5, Prop. 4.5], this can happen only if $\dim \mathcal{H}_\lambda < \infty$.

1. and 2. We prove that $\sigma_p(T)$ is non empty. If so it is certainly real since T is per assumption self-adjoint and all its eigenvalues are real as per item 2. of Corollary 38. We can rewrite (9) as

$$-\|T\| \leq (\psi, T\psi) \leq \|T\|, \quad \forall \psi \in \mathcal{H} \text{ such that } \|\psi\| = 1$$

which entails that the equality can be obtained either as the infimum or as the supremum of $(\psi, T\psi)$ over all unit norm $\psi \in \mathcal{H}$. Consider the latter case, the former following suit. Assume $\|T\| > 0$ or the theorem becomes trivial as $T = 0$. Fix an eigenvalue λ and a relative eigenvector ψ_λ such that $\|\psi_\lambda\| = 1$. The Cauchy-Schwarz inequality entails that

$$\|T\| \geq |(\psi_\lambda, T\psi_\lambda)| = |\lambda|(\psi_\lambda, \psi_\lambda) = |\lambda|.$$

In other words $\sup_{\lambda \in \sigma_p(T)} \{|\lambda|\} \leq \|T\|$. We exhibit now an eigenvector whose eigenvalue is indeed $\|T\|$. Consider a sequence of elements $\{\psi_n\}_{n \in \mathbb{N}}$ such that $\|\psi_n\| = 1$ for all n and $(\psi_n, T\psi_n) \rightarrow \|T\|$ as $n \rightarrow \infty$. Since $\|T(\psi_n)\| \leq \|T\| \|\psi_n\| = \|T\|$, it holds

$$\|(T(\psi_n) - \|T\| \psi_n)\|^2 = \|T(\psi_n)\|^2 + \|T\|^2 - 2\|T\|(\psi_n, T\psi_n) \leq 2\|T\|^2 - 2\|T\|(\psi_n, T\psi_n),$$

which converges to 0 as $n \rightarrow \infty$. To conclude we need to extract from $\{\psi_n\}_{n \in \mathbb{N}}$ a converging subsequence. Since the sequence is bounded, being $\|\psi_n\| = 1$ for all $n \in \mathbb{N}$, and since T is compact per assumption, there exists $\{T(\psi_{n_k})\}_{n_k \in \mathbb{N}}$, a converging subsequence of $\{T(\psi_n)\}_{n \in \mathbb{N}}$. As a by-product, define

$$\psi_{n_k} = \frac{1}{\|T\|} (T(\psi_{n_k}) - (T(\psi_{n_k}) - \|T\| \psi_{n_k})),$$

which converges to a certain $\psi \in \mathcal{H}$ as $n_k \rightarrow \infty$ since $\lim_{n_k \rightarrow \infty} (T(\psi_{n_k}) - \|T\| \psi_{n_k}) = 0$. In other words we have shown that $T\psi = \|T\|\psi$, which is the sought conclusion.

3. This is straightforward once we established that if 0 is the only eigenvalue, then $\|T\| = 0$, hence $T = 0$. $\square$

The more celebrated theorem proven by Hilbert is actually a continuation of the previous one:

Theorem 62 (Hilbert). *Let $\mathcal{H}$ be an Hilbert space and let T be a self-adjoint, compact operator thereon and let $\sigma_p(T)$ be the collection of its eigenvalues. Then*

1. *If P_λ is the orthogonal projector on the eigenspace of T relative to the eigenvalue λ , we can write*

$$T = \sum_{\lambda \in \sigma_p(T)} \lambda P_\lambda,$$

where the sum is understood in the uniform topology and all eigenvalues are collected in a decreasing order with respect to their absolute value.

2. *$\mathcal{H}$ admits a basis of eigenvectors of T .*

Proof. 1. Consider any eigenvalue $\lambda \in \mathbb{C}$ with eigenspace $\mathcal{H}_\lambda$ and let $P_\lambda : \mathcal{H} \rightarrow \mathcal{H}_\lambda$ be the orthogonal projector. Theorem 42 guarantees us that $\mathcal{H} = \mathcal{H}_\lambda \oplus \mathcal{H}_\lambda^\perp$ and that $Q_\lambda = \mathbb{I} - P_\lambda$ is an orthogonal projector on $\mathcal{H}_\lambda^\perp$. In addition we can show that

$$TP_\lambda = P_\lambda T = \lambda P_\lambda.$$

As a matter of fact, for any $\psi \in \mathcal{H}$, it holds

$$TP_\lambda \psi = \lambda P_\lambda \psi \implies TP_\lambda = \lambda P_\lambda$$

By taking the adjoint and using the hypothesis that $P_\lambda^* = P_\lambda$ and $T^* = T$, we obtain

$$(TP_\lambda)^* = (\lambda P_\lambda)^* \iff P_\lambda T = \lambda P_\lambda,$$

where we used that $\lambda \in \mathbb{R}$ as per Theorem 61. Following slavishly this line of reasoning we draw the same conclusion with respect to Q_λ , namely $Q_\lambda T = TQ_\lambda$. It holds

$$T = T\mathbb{I} = T(P_\lambda + Q_\lambda) = \lambda P_\lambda + Q_\lambda T.$$

In view of the properties of both Q_λ and T we know that $(Q_\lambda T)^* = TQ_\lambda = Q_\lambda T$, that is $Q_\lambda T$ is self-adjoint. Furthermore it is compact since $T \in \mathcal{B}_\infty(\mathcal{H})$ and all projectors Q_λ are bounded. In addition, since $P_\lambda Q_\lambda = Q_\lambda P_\lambda$, it holds

$$P_\lambda(Q_\lambda T) = (Q_\lambda T)P_\lambda.$$

In view of these results, consider $\lambda_0 \in \mathbb{C}$ the eigenvalue of T with highest absolute value. Using the notation $\mathcal{H}_0 \equiv \mathcal{H}_{\lambda_0}$ and similarly for the projection operators, we write $T = \lambda_0 P_0 + T_1$, where $T_1 = Q_0 T$. If $T_1 = 0$ the proof ends; if not, since $T_1 \in \mathcal{B}_\infty(\mathcal{H})$ and since it is self-adjoint, we consider λ_1 , the eigenvalue of T_1 with highest absolute value. Let ψ_1 be an eigenvector relative to λ_1 . This is also an eigenvector of T since

$$T\psi_1 = \lambda_0 P_0 \psi_1 + T_1 \psi_1 = \lambda_1 \psi_1,$$

since $\psi \in \mathcal{H}_0^\perp$. Hence the eigenspace of T and of T_1 relative to λ_1 coincide and we can write $T = \lambda_0 P_0 + \lambda_1 P_1 + T_2$. From Theorem 61 we know, moreover, that $\|T\| = \lambda_0$ and $\|T_1\| = \lambda_1$. We can reiterate the construction for as many steps as necessary obtaining

$$T = \sum_{i=0}^{n-1} \lambda_i P_i + T_n,$$

where $|\lambda_0| \geq |\lambda_1| \geq \dots \geq |\lambda_{n-1}|$. We claim that this sequence of eigenvalues must converge to 0. To show it, consider a vector of unit norm ψ_i in each of the eigenspace $\mathcal{H}_i$. This is a bounded sequence and, thus, being T compact per hypothesis, there must exist a subsequence of $\{T(\psi_i)\}_{i \in \mathbb{N}}$ converging. Suppose by contradiction that there exists $\epsilon > 0$ such that the absolute value of all eigenvalues is greater than ϵ . Then, for all p, m with $p > m$

$$\|T(\psi_p) - T(\psi_m)\|^2 = \|\lambda_p \psi_p - \lambda_m \psi_m\|^2 = |\lambda_p|^2 + |\lambda_m|^2 \geq 2\epsilon,$$

where we used in the last equality that, per construction ψ_p and ψ_m are orthogonal. The above inequality entails that $\lambda_n \rightarrow 0$ as $n \rightarrow \infty$. Hence $\{T(\psi_n)\}_{n \in \mathbb{N}}$ is not a Cauchy sequence and we cannot extract from it a converging subsequence. In other words we have shown that

$$T = \sum_{n=0}^{\infty} \lambda_n P_n,$$

which converges in the uniform topology. From the construction it is also clear that the ordering of the eigenvalues is dictated by convenience and ultimately it plays no role.

We need to prove that the sum encompasses all non-vanishing eigenvalues of T . Suppose there exists $\lambda \neq \lambda_n$ for all possible values of n . Let us call P_λ the orthogonal projective on the associated eigenspace. Since eigenspaces relative to distinct eigenvalues of a self-adjoint, bounded operator are orthogonal, *c.f.* Proposition 37, it holds that $P_\lambda P_{\lambda_n} = P_{\lambda_n} P_\lambda = 0$ for all possible values of n . In addition

$$TP_\lambda = \sum_{n=0}^{\infty} \lambda_n P_{\lambda_n} P_\lambda = 0,$$

which entails that, for every $\psi_\lambda \in \mathcal{H}_\lambda$ $T\psi_\lambda = TP_\lambda \psi_\lambda = 0$, namely ψ_λ is an eigenvector of eigenvalue 0. In other words $\lambda = 0$.

2. Consider each eigenspace $\mathcal{H}_n$ and consider an orthogonal basis on each of these spaces. We need to prove that the union of all these basis is a set of generators for $\mathcal{H}$. It suffices that the orthogonal complement of their span is trivial. Let ψ be in such orthogonal space. Then $T\psi = \sum_{n=0}^{\infty} \lambda_n P_{\lambda_n} \psi = 0$. In other words $\psi \in \ker(T)$, namely it is an eigenvector of eigenvalue 0. At the same time, being orthogonal to each of the eigenspaces built from each P_{λ_n} , $\psi \in \ker(T)^\perp$. To summarize $\psi \in \ker(T) \cap \ker(T)^\perp = \{0\}$. $\square$

Finished? Nope, we have barely begun!!! But we learned something: knowing the eigenvalues of a compact, self-adjoint operator together with their multiplicities is tantamount to knowing the whole operator. Isn't it like with matrices? I told you so! Oh, before I forget, I wanted to tell you (probably "you" is just me ... ah no, ciao Beatrice!) one last thing without proving it:

Theorem 63. *Let $\mathcal{H}$ be a Hilbert space and let $A \in \mathcal{B}_\infty(\mathcal{H})$ with $A \neq 0$. Let $\lambda_i, i \in \mathbb{N} \cup \{0\}$ be the non-vanishing eigenvalues of $|A|$ ordered decreasingly. They are called $\text{sing}(A)$, the **singular values** of A . Let $\psi_{\lambda_i, j} \in \mathcal{H}$ be an orthonormal basis of the eigenspace relative to λ_i with j running from 1 to m_i the multiplicity of λ_i . Then*

$$A(\cdot) = \sum_{\lambda_i \in \text{sing}(A)} \sum_{j=1}^{m_i} \lambda_i (\psi_{\lambda_i, j}, \cdot) \tilde{\psi}_{\lambda_i, j},$$

and

$$A^*(\cdot) = \sum_{\lambda_i \in \text{sing}(A)} \sum_{j=1}^{m_i} \lambda_i (\tilde{\psi}_{\lambda_i, j}, \cdot) \psi_{\lambda_i, j},$$

where $\tilde{\psi}_{\lambda_i, j} \doteq \frac{1}{\lambda_i} A(\psi_{\lambda_i, j})$.

3.3 Hilbert-Schmidt Operators

We devote this section to introducing and to studying the main properties of a notable class of compact operators with manifold applications ranging from quantum mechanics to theory of PDEs.

Definition 64. *Let $\mathcal{H}$ be a Hilbert space. We say that $A \in \mathcal{B}(\mathcal{H})$ is a **Hilbert-Schmidt operator** if there is an orthonormal basis $\{\psi_i\}_{i \in \mathbb{N}}$ in $\mathcal{H}$ such that*

$$\sum_{i=1}^{\infty} \|A\psi_i\|^2 < \infty.$$

We call $\mathcal{B}_2(\mathcal{H})$ the collection of all Hilbert-Schmidt operators on $\mathcal{H}$ and we call

$$\|A\|_2 \doteq \sqrt{\sum_{i=1}^{\infty} \|A\psi_i\|^2}. \quad (12)$$

One faces at least two dilemmas: Is the definition independent from the choice of a basis and why is this class of operators so useful? We shall give immediately an answer to both queries (otherwise you might have a sleepless night!). Before proceeding, recall that we are assuming that all Hilbert spaces are separable.

Theorem 65. *Let $\mathcal{H}$ be a Hilbert space and let $\{\psi_i\}_{i \in \mathbb{N}}, \{\phi_j\}_{j \in \mathbb{N}}$ be two orthonormal bases of $\mathcal{H}$. Let $A \in \mathcal{B}(\mathcal{H})$; then*

(a) $\sum_{i=1}^{\infty} \|A\psi_i\|^2$ is finite if and only if such is $\sum_{j=1}^{\infty} \|A\phi_j\|^2$ and the two sums coincide,

(b) $\sum_{i=1}^{\infty} \|A\psi_i\|^2$ is finite if and only if such is $\sum_{j=1}^{\infty} \|A^*\phi_j\|^2$ and the two sums coincide.

Proof. Let us consider an arbitrary $A \in \mathcal{B}(\mathcal{H})$ and two bases of $\mathcal{H}$ as per hypothesis. Assume that $\|A\|_2^2$ is finite; then the following chain of equalities holds in view of Parseval identity:

$$\sum_{i=1}^{\infty} \|A(\psi_i)\|^2 = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} |(A(\psi_i), \phi_j)|^2 = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} |(\psi_i, A^*(\phi_j))|^2 = \sum_{j=1}^{\infty} \|A^*(\phi_j)\|^2,$$

where the two sums can be exchanged since all terms are positive. All these identities would hold true having assumed that the right most term would be convergent. In other words we have proven item (b). The first one is now a direct consequence of (b). $\square$

The last theorem guarantees that the notion of an Hilbert-Schmidt operator is well-posed, but it does not tell us why it is useful. Well, the reason is ultimately that $\mathcal{B}_2(\mathcal{H})$ is a Hilbert space. The following proposition clarifies better this statement.

Proposition 66. *Let $\mathcal{H}$ be a Hilbert space. Then $\mathcal{B}_2(\mathcal{H})$ is a subspace of $\mathcal{B}(\mathcal{H})$ such that*

1. $\|A\|_2 = \|A^*\|_2$ for all $A \in \mathcal{B}_2(\mathcal{H})$,
2. $\|AK\|_2 \leq \|K\| \|A\|_2$ and $\|KA\|_2 \leq \|K\| \|A\|_2$ for all $A \in \mathcal{B}_2(\mathcal{H})$ and for all $K \in \mathcal{B}(\mathcal{H})$,
3. $\|A\| \leq \|A\|_2$ for all $A \in \mathcal{B}_2(\mathcal{H})$.

Proof. To start with, we need to prove that $\mathcal{B}_2(\mathcal{H})$ is a linear subspace of $\mathcal{B}(\mathcal{H})$. It is per construction closed under multiplication by a scalar. We need only to verify that, for all $A, B \in \mathcal{B}_2(\mathcal{H})$, $(A + B) \in \mathcal{B}_2(\mathcal{H})$. Let $\{\psi_i\}_{i \in \mathbb{N}}$ be an arbitrary orthonormal basis. Then

$$\sum_{i=1}^{\infty} \|(A + B)\psi_i\|^2 \leq \sum_{i=1}^{\infty} (\|A\psi_i\| + \|B\psi_i\|)^2 \leq 2 \left[\sum_{i=1}^{\infty} \|B\psi_i\|^2 + \sum_{i=1}^{\infty} \|A\psi_i\|^2 \right] < \infty,$$

where we used the triangle inequality.

1. is a direct consequence of the proof of Theorem 65.
2. Let $A \in \mathcal{B}_2(\mathcal{H})$, $K \in \mathcal{B}(\mathcal{H})$ and let $\{\psi_i\}_{i \in \mathbb{N}}$ be an arbitrary orthonormal basis. Then

$$\|KA\|_2^2 = \sum_{i=1}^{\infty} \|(KA)\psi_i\|^2 \leq \sum_{i=1}^{\infty} \|K\|^2 \|A\psi_i\|^2 = \|K\|^2 \|A\|_2^2.$$

In other words $KA \in \mathcal{B}_2(\mathcal{H})$. At the same time, by item 1. we know that

$$\|(AK)^*\|_2 = \|K^*A^*\|_2 \leq \|K^*\| \|A^*\|_2 = \|K\| \|A\|_2,$$

where we used Theorem 65 to conclude that $A^* \in \mathcal{B}_2(\mathcal{H})$ as well as Proposition 32 to infer that $\|K^*\| = \|K\|$.

3. Let $A \in \mathcal{B}_2(\mathcal{H})$ and let $\{\psi_i\}_{i \in \mathbb{N}}$ be an arbitrary orthonormal basis. It holds by (7)

$$\begin{aligned} \|A\| &= \sup_{\phi \in \mathcal{H} \mid \|\phi\|=1} \|A\phi\| = \sup_{\phi \in \mathcal{H} \mid \|\phi\|=1} (\|A\phi\|^2)^{\frac{1}{2}} = \\ &= \sup_{\phi \in \mathcal{H} \mid \|\phi\|=1} \left(\sum_{i=1}^{\infty} |(\psi_i, A\phi)|^2 \right) = \sup_{\phi \in \mathcal{H} \mid \|\phi\|=1} \left(\sum_{i=1}^{\infty} |(A^*\psi_i, \phi)|^2 \right)^{\frac{1}{2}} \leq \\ &\leq \sup_{\phi \in \mathcal{H} \mid \|\phi\|=1} \left(\sum_{i=1}^{\infty} \|A^*\psi_i\|^2 \|\phi\|^2 \right)^{\frac{1}{2}} = \|A^*\|_2 = \|A\|_2, \end{aligned}$$

where we used at different stages both the Parseval equality and item 1. $\square$

From a more technical perspective this proposition tell us that Hilbert-Schmidt operators not only form an algebra but rather an ideal in the algebra of bounded operators.

Proposition 67. *Let $\mathcal{H}$ be an Hilbert space and let $A, B \in \mathcal{B}_2(\mathcal{H})$. Let $\{\psi_i\}_{i \in \mathbb{N}}$ be an orthonormal basis of $\mathcal{H}$. Let $(\cdot, \cdot)_2 : \mathcal{B}_2(\mathcal{H}) \times \mathcal{B}_2(\mathcal{H}) \rightarrow \mathbb{C}$ be*

$$(A, B)_2 \doteq \sum_{i=1}^{\infty} (A\psi_i, B\psi_i). \quad \forall A, B \in \mathcal{B}_2(\mathcal{H}) \quad (13)$$

This is well-defined map which determines a scalar-product on $\mathcal{B}_2(\mathcal{H})$ such that, for all $A \in \mathcal{B}_2(\mathcal{H})$, $\|A\|_2^2 = (A, A)_2$.

Proof. Notice that the map $(\cdot, \cdot)_2$ is bilinear and convergent, since, for any $i \in \mathbb{N}$

$$\sum_{i=1}^{\infty} |(A\psi_i, B\psi_i)| \leq \sum_{i=1}^{\infty} \|A\psi_i\| \|B\psi_i\| \leq \frac{1}{2} \sum_{i=1}^{\infty} (\|A\psi_i\|^2 + \|B\psi_i\|^2) \leq \infty,$$

for any choice of $A, B \in \mathcal{B}_2(\mathcal{H})$. The last chain of inequalities holds true due to the Cauchy-Schwarz inequality. Having established convergence, it is straightforward to verify that we have constructed an Hermitian inner product as per Definition 9 since all properties are inherited from those of $(\cdot, \cdot)$. To conclude we observe that, once it has been established that $(\cdot, \cdot)_2$ is an Hermitian inner product, we have the polarisation identity at our disposal which, in combination with Theorem 65, entails that $(\cdot, \cdot)_2$ does not depend on the choice of the orthonormal basis $\{\psi_i\}_{i \in \mathbb{N}}$. $\square$

We conclude this section with two additional results, starting from a corollary whose proof we omit.

Corollary 68. *Let $\mathcal{H}$ be a Hilbert space. Then every Hilbert-Schmidt operator is also compact.*

Proposition 69. *Let $\mathcal{H}$ be an Hilbert space. Then $\mathcal{B}_2(\mathcal{H})$ is a Hilbert space with respect to (13). Furthermore, for every $A \in \mathcal{B}_2(\mathcal{H})$*

$$\|A\|_2^2 = \sum_{\lambda \in \text{sing}(A)} m_\lambda \lambda^2,$$

where m_λ is the multiplicity of the eigenvalue λ of $|A|$.

Proof. We divide our analysis in two parts. In the first we show that $\mathcal{B}_2(\mathcal{H})$ is a Hilbert space, in the second we prove the remaining statements of the proposition.

In view of Proposition 67 we need to show only the completeness of $\mathcal{B}_2(\mathcal{H})$. Consider an arbitrary orthonormal basis $\{\psi_i\}_{i \in \mathbb{N}}$ and let $\{A_n\}_{n \in \mathbb{N}}$ be a Cauchy sequence of Hilbert-Schmidt operators. In view of item 3. of Proposition 66 we know that $\{A_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence in $\mathcal{B}(\mathcal{H})$. The latter is a Banach space, hence complete. There exists $A \in \mathcal{B}(\mathcal{H})$ such that $\lim_{n \rightarrow \infty} \|A_n - A\| = 0$. By definition of Cauchy sequence, we know that for all $\epsilon > 0$, we can find two sufficiently large indices $m, k \in \mathbb{N}$ such that

$$\sum_{i=1}^{\infty} \|(A_k - A_m)\psi_i\|^2 = \|A_k - A_m\|_2^2 \leq \epsilon^2.$$

Hence for every but fixed $q \in \mathbb{N}$

$$\sum_{i=1}^q \|(A_k - A_m)\psi_i\|^2 \leq \epsilon^2,$$

which allows us to take the limit as $m \rightarrow \infty$ to obtain

$$\sum_{i=1}^q \|(A_k - A)\psi_i\|^2 \leq \epsilon^2.$$

In view of the arbitrariness of q , it descends

$$\sum_{i=1}^{\infty} \|(A_k - A)\psi_i\|^2 \leq \epsilon^2,$$

which entails that $(A - A_k) \in \mathcal{B}_2(\mathcal{H})$. Since $A_k \in \mathcal{B}_2(\mathcal{H})$ per hypothesis, we have that $A = (A - A_k) + A_k \in \mathcal{B}_2(\mathcal{H})$ thanks to Proposition 66. We have shown the convergence of the Cauchy sequence to an element of $\mathcal{B}_2(\mathcal{H})$.

Consider now $A \in \mathcal{B}_2(\mathcal{H})$ and $|A|$. Since Corollary 68 entails that $A \in \mathcal{B}_\infty(\mathcal{H})$, then $|A|$ is both positive and compact, see Definition 47 and Proposition 59. Following Theorem 63, we construct first a basis $\{\phi_{\lambda,i}\}_{\lambda \in \text{sing}(A)}$ of $(\ker(|A|))^\perp$ where each eigenvector comes with a

multiplicity i running from 1 to m_λ . We complete the basis of $\mathcal{H}$ with $\{\psi_j\}_{j \in \mathbb{N}}$, a basis of $\ker(|A|)$ which coincides with $\ker(A)$ by Corollary 48. We can thus write

$$\|A\|_2^2 = \sum_{\lambda \in \text{sing}(A)} \sum_{i=1}^{m_\lambda} (A\phi_{\lambda,i}, A\phi_{\lambda,i}) = \sum_{\lambda \in \text{sing}(A)} \sum_{i=1}^{m_\lambda} (A^*A\phi_{\lambda,i}, \phi_{\lambda,i}) = \sum_{\lambda \in \text{sing}(A)} \lambda^2 m_\lambda,$$

where we used that $|A|\phi_{\lambda,i} = \sqrt{A^*A}\phi_{\lambda,i} = \lambda\phi_{\lambda,i}$. $\square$

3.4 Trace Class Operators

In this section we define and we discuss the main properties of a notable class of bounded operators, which are known either as trace class or as **nuclear operators**. Their relevance is enormous being naturally associated to quantum states in quantum mechanics over finite dimensional Hilbert spaces and being the starting point for the notion of thermal states in quantum statistical mechanics. As in the previous section, we will be assuming that all Hilbert spaces are separable even though we shall not mention it explicitly and, for this reason, all calculations will be done under the assumption that the Hilbert space is infinite dimensional. If this were not to be the case, all results would still be true barring the change of a rather limited (and irrelevant) number of details. As a starting point we need a preliminary result:

Proposition 70. *Let $\mathcal{H}$ be a Hilbert space and let $A \in \mathcal{B}(\mathcal{H})$. The following three statements are equivalent:*

1. *There exists an orthonormal basis $\{\psi_i\}_{i \in \mathbb{N}}$ such that*

$$\sum_{i=1}^{\infty} (\psi_i, |A|\psi_i) < \infty.$$

2. *$\sqrt{|A|}$ is a Hilbert-Schmidt operator,*

3. *$A \in \mathcal{B}_\infty(\mathcal{H})$ and the sum of all elements of the set $\{\lambda m_\lambda\}_{\lambda \in \text{sing}(A)}$ is finite.*

Proof. 1. $\iff$ 2. Since $|A| = \sqrt{|A|}\sqrt{|A|}$ and since both $|A|$ and $\sqrt{|A|}$ are self-adjoint per construction, condition 1. translates as

$$\sum_{i=1}^{\infty} (\psi_i, \sqrt{|A|}\sqrt{|A|}\psi_i) = \sum_{i=1}^{\infty} (\sqrt{|A|}\psi_i, \sqrt{|A|}\psi_i) = \|\sqrt{|A|}\|_2^2 < \infty,$$

which entails that $\sqrt{|A|} \in \mathcal{B}_2(\mathcal{H})$. The converse follows by tracing backwards this reasoning.

2. $\iff$ 3. Corollary 68 entails that $\sqrt{|A|}$ is a compact operator. Proposition 56 guarantees that the product of compact operators is still compact, hence $|A| \in \mathcal{B}_\infty(\mathcal{H})$. Let us consider

the basis made out of unit norm eigenvectors of $|A|$, namely $\{\psi_{\lambda,i}\}$, i running from 1 to m_λ , the multiplicity of λ . Since $|A|\psi_{\lambda,i} = \lambda\psi_{\lambda,i}$ for any value of i , it holds

$$\|\sqrt{A}\|_2^2 = \sum_{\lambda} \sum_{i=1}^{m_\lambda} \left(\sqrt{|A|}\psi_{\lambda,i}, \sqrt{|A|}\psi_{\lambda,i} \right) = \sum_{\lambda} \sum_{i=1}^{m_\lambda} (\psi_{\lambda,i}, |A|\psi_{\lambda,i}) = \sum_{\lambda \in \text{sing}(A)} \lambda m_\lambda,$$

where in the second equality we used that $\sqrt{|A|}$ is self-adjoint and in the last we observed that all eigenvectors of the eigenvalue 0 do not contribute to the sum. Tracing backwards the proof shows that also the converse implication holds true. $\square$

In view of this last proposition,

Definition 71. Let $\mathcal{H}$ be a Hilbert space and let $A \in \mathcal{B}(\mathcal{H})$. We call it a **trace class operator** if $\sqrt{|A|}$ is a Hilbert-Schmidt operator. The set of all these operators is indicated as $\mathcal{B}_1(\mathcal{H})$ and, for all $A \in \mathcal{B}_1(\mathcal{H})$,

$$\|A\|_1 \doteq \|\sqrt{|A|}\|_2^2 = \sum_{\lambda \in \text{sing}(A)} \lambda m_\lambda.$$

We have now defined different classes of operators on Hilbert spaces, but we have not yet completely investigated their relation. The following corollary goes one step in this direction:

Corollary 72. Let $\mathcal{H}$ be a Hilbert space. The following inclusions hold

$$\mathcal{B}_1(\mathcal{H}) \subset \mathcal{B}_2(\mathcal{H}) \subset \mathcal{B}_\infty(\mathcal{H}) \subset \mathcal{B}(\mathcal{H}).$$

Proof. We have already shown all the inclusions except for the first one. Suppose $A \in \mathcal{B}_1(\mathcal{H})$. Per Definition 71, this means that $\sqrt{|A|} \in \mathcal{B}_2(\mathcal{H})$, but item 2. of Proposition 66 entails that the product of Hilbert-Schmidt operators lies still in this class. Hence $|A| = \sqrt{|A|}\sqrt{|A|} \in \mathcal{B}_2(\mathcal{H})$. Since A is bounded per assumption, we can use the polar decomposition of Theorem 49 to write $A = U|A|$ where $U \in \mathcal{B}(\mathcal{H})$. Still item 2. of Proposition 66 guarantees that $A \in \mathcal{B}_2(\mathcal{H})$. $\square$

I am sure that, at the fifth semester of your studies, you have learned that no name is give by chance to an important mathematical object. Hence you must (and I stress that you must, especially you Simone!) be wondering why the name trace class and how an operation, which is intrinsically well-defined on a finite dimensional Hilbert space, can be generalized to the infinite dimensional scenario. Before answering to this question, time for a filler episode... well for another theorem which, in a mathematically advanced language, is telling us that trace-class operators form an ideal in the space of bounded operators. As we have already proven several similar results, we shall omit the proof, which can be found in [5, Theorem 4.32]:

Theorem 73. Let $\mathcal{H}$ be a Hilbert space. The following statements hold true:

- (a) For every $A \in \mathcal{B}_1(\mathcal{H})$, there exist $B, C \in \mathcal{B}_2(\mathcal{H})$ such that $A = BC$. Conversely for every pair of Hilbert-Schmidt operators B, C on $\mathcal{H}$, $BC \in \mathcal{B}_1(\mathcal{H})$ and $\|BC\|_1 \leq \|B\|_2 \|C\|_2$.

(b) $\mathcal{B}_1(\mathcal{H})$ is a linear subspace of $\mathcal{B}(\mathcal{H})$ and, moreover, $\forall A \in \mathcal{B}_1(\mathcal{H})$ and $\forall K \in \mathcal{B}(\mathcal{H})$, $KA, AK \in \mathcal{B}_1(\mathcal{H})$ and

$$\|KA\|_1 \leq \|K\| \|A\|_1 \quad \text{and} \quad \|AK\|_1 \leq \|K\| \|A\|_1,$$

(c) $\|\cdot\|_1$ is a norm on $\mathcal{B}_1(\mathcal{H})$.

One could go one step further actually showing that $\mathcal{B}_1(\mathcal{H})$ is complete with respect to the norm $\|\cdot\|_1$, hence it is a Banach space.

We can now answer to question asked above:

Proposition 74. *Let $\mathcal{H}$ be a Hilbert space. If $\{\psi_i\}_{i \in \mathbb{N}}$ is an orthonormal basis of $\mathcal{H}$, we call **trace** $Tr : \mathcal{B}_1(\mathcal{H}) \rightarrow \mathbb{C}$ the map*

$$Tr(A) = \sum_{i=1}^{\infty} (\psi_i, A\psi_i). \quad \forall A \in \mathcal{B}_1(\mathcal{H}) \quad (14)$$

In addition it holds that

1. The image of Tr does not depend on the choice of the basis,
2. For every $B, C \in \mathcal{B}_2(\mathcal{H})$, it holds

$$Tr(BC) = (B^*, C)_2,$$

3. for every $A \in \mathcal{B}_1(\mathcal{H})$, $|A| \in \mathcal{B}_1(\mathcal{H})$ and

$$\|A\|_1 = Tr(|A|).$$

Proof. 1. and 2. In view of Theorem 73 we can write any $A \in \mathcal{B}_1(\mathcal{H})$ as the product $A = BC$, $B, C \in \mathcal{B}_2(\mathcal{H})$. In view of (13), we can write

$$(B^*, C)_2 = \sum_{i=1}^{\infty} (B^* \psi_i, C \psi_i) = \sum_{i=1}^{\infty} (\psi_i, BC \psi_i) = \sum_{i=1}^{\infty} (\psi_i, A \psi_i) = Tr(A),$$

where $\{\psi_i\}_{i \in \mathbb{N}}$ is an orthonormal basis of $\mathcal{H}$. Since item 1. of Proposition 73 guarantees us also the the product of two Hilbert-Schmidt operators is trace class, this last chain of identities tell us not only that 2. but also that 1. is true, since the Hilbert-Schmidt Hermitian inner product does not depend on the choice of the basis.

3. Since $|A| = (|A|)$ we know that from (12) $\| |A| \|_1 = \|A\|_1$ which is finite since $A \in \mathcal{B}_1(\mathcal{H})$ per assumption. If we choose $\{\psi_{\lambda,i}\}$ a basis of eigenvectors of $|A|$, with the index i running from 1 to m_λ , the multiplicity of the eigenvalue λ , it holds per Definition 71 that

$$Tr(|A|) = \sum_{\lambda \in \text{sing}(A)} \sum_{i=1}^{m_\lambda} (\psi_{\lambda,i}, |A| \psi_{\lambda,i}) = \sum_{\lambda \in \text{sing}(A)} \sum_{i=1}^{m_\lambda} \lambda = \sum_{\lambda \in \text{sing}(A)} \lambda m_\lambda = \|A\|_1,$$

which is the sought identity. $\square$

This last proposition is indeed a justification to the use of the word trace, but one might still be wondering whether the operator Tr inherits also all the properties of the trace of a matrix. The answer is essentially yes, but it does not hurt (me, you... maybe!) to prove some of them in this context:

Proposition 75. *Let $\mathcal{H}$ be a Hilbert space. Then the trace operator enjoys the following properties:*

- For every $A, B \in \mathcal{B}_1(\mathcal{H})$ and for all $\alpha, \beta \in \mathbb{C}$

$$Tr(\alpha A + \beta B) = \alpha Tr(A) + \beta Tr(B), \quad Tr(A^*) = \overline{Tr(A)},$$

- it is cyclic, namely, for every $A \in \mathcal{B}_1(\mathcal{H})$ and for every $K \in \mathcal{B}(\mathcal{H})$, it holds

$$Tr(AK) = Tr(KA).$$

Proof. 1. It is an immediate consequence of (14) and of the bi-linearity of the Hermitian inner product on $\mathcal{H}$.

2. To start with, suppose that both A and K lie in $\mathcal{B}_2(\mathcal{H})$. Then item 2. of Proposition 74 guarantees that

$$Tr(AK) = (A^*, K)_2 \quad \text{and} \quad Tr(KA) = (K^*, A)_2.$$

Since $\mathcal{B}_2(\mathcal{H})$ is a Hilbert space with respect to $(\cdot, \cdot)_2$, we can use the polarisation identity to write

$$4(A^*, K)_2 = \|A^* + K\|_2^2 + \|A^* - K\|_2^2 - i\|A^* + iK\|_2^2 + i\|A^* - iK\|_2^2.$$

Since item 1. in Proposition 66 guarantees us, that for every $C \in \mathcal{B}_2(\mathcal{H})$, $\|C^*\|_2 = \|C\|_2$, we can rewrite the above identity as

$$4(A^*, K)_2 = \|A + K^*\|_2^2 + \|A - K^*\|_2^2 - i\|K^* + iA\|_2^2 + i\|K^* - iA\|_2^2 = 4(K^*, A).$$

Suppose now that $K \in \mathcal{B}(\mathcal{H})$, but not necessarily compact. In view of item 1. of Theorem 73, we can decompose $A = CD$, with $C, D \in \mathcal{B}_2(\mathcal{H})$. Hence

$$Tr(AK) = Tr(CDK) = Tr(DKC) = Tr(KCD) = Tr(KA),$$

which concludes the proof. □

4 Unbounded Operators

In the previous section we have studied bounded operators and we have seen that their behaviour is well-under control. So where is the problem? Well... most of the physically interesting operators do not fall in that class. The prime examples are the position and the momentum operators, not to mention the various Hamiltonians which are at the hearth of manifold models.

Depending on the depth of your knowledge of quantum mechanics, the above examples might or might not make a lot of sense. In any case let me offer you, my dear reader(s), an example of what I mean.

Consider $\mathcal{H} = L^2(I)$ where $I = (1, \infty)$ and the measure is the standard Lebesgue one. A natural operation that we want to be able to perform on such Hilbert space is the multiplication by x the coordinate along $I \subset \mathbb{R}$. This is a special example of the position operator that we will be studying at a later stage. Yet the question is whether what we are trying to define is meaningful. Consider two cases:

- Let $\psi(x) = e^{-\frac{x}{2}}$. This lies in $\mathcal{H}$ since

$$\int_1^\infty dx |\psi(x)|^2 = \int_1^\infty dx e^{-x} = -e^{-x}|_1^\infty = \frac{1}{e}.$$

Multiplying $\psi(x)$ by x yields $xe^{-\frac{x}{2}}$ and

$$\int_1^\infty dx x^2 e^{-x} = \frac{5}{e},$$

which guarantees us that $x\psi(x) \in \mathcal{H}$.

- Let $\psi(x) = x^{-1}$. This lies in $\mathcal{H}$ since

$$\int_1^\infty dx x^{-2} = -x^{-1}|_1^\infty = 1.$$

Multiplying by x yields the constant function 1, whose integral from 1 to infinity clearly diverges.

In other words it is easy to define a large class of operators which are well-defined only on certain elements of a given Hilbert spaces. As we will see in this section, especially via the Hillinger-Toeplitz theorem, these operators are characterized by one notable characteristic: They are unbounded, namely their operator norm as per (7) diverges. As before, we shall try to keep the minimum number of assumptions and structures.

Hence as a starting point we consider only complex vector spaces V . As a rule, from now on, we have in mind the infinite dimensional scenario:

Definition 76. *Let V be a complex vector space. We call a linear map*

$$T : D(T) \subset V \rightarrow V,$$

*an **operator** on the **domain** $D(T)$. The **graph** of T is defined as*

$$G(T) = \{(v, T(v)) \in V \oplus V \mid v \in D(T)\}.$$

As a straightforward consequence of this definition, we can infer that the standard operations between operators are still meaningful provided we pay attention at the correct domain of definition. As a matter of fact:

- if T, S are two operators on V , then also TS is an operator on V with the so-called **standard domain**

$$D(TS) = \{v \in D(S) \mid S(v) \in D(T)\},$$

- if T is an operator on V and $\alpha \in \mathbb{C}$, also αT is an operator on V with **standard domain** $D(\alpha T) = D(T)$, if $\alpha \neq 0$. If $\alpha = 0$, then the operator is everywhere defined coinciding with the trivial operator.

Notice that, in view of these remarks, the associative and distributive properties respectively of the sum and of the product of operators are valid. Furthermore it happens in many concrete cases that we are able to define easily an operator on a given domain, to realize at a later stage that such domain could have been enlarged. This leads to the following rather intuitive definition:

Definition 77. Let V be a vector space and let T be an operator thereon. A second operator T' is called an **extension** of T if $G(T) \subset G(T')$. In this case we write $T \subset T'$.

We consider now a normed space X in place of a simple vector space. Observe that the Cartesian product $X \times X$ is at the same time the vector space $X \oplus X$ and a topological space endowed with the product topology as per Definition 212. This is important since it makes precise the notion of a closed set and of a closure in $X \times X$, hence justifying the following definition which introduce a key concept in the theory of operators:

Definition 78. Let X be a normed space and let $A : D(A) \subseteq X \rightarrow X$ be an operator. We say that

- A is **closed** if $G(A)$ is closed in the product topology – see Definition 212,
- A is **closable** if $\overline{G(A)}$ is the graph of an operator, which we denote as $\bar{A}$, the **closure** of A .

Exercise 15. Show that an equivalent definition of a closed operator is the following: A is closed if and only if for every sequence $\{x_n\}_{n \in \mathbb{N}}$, $x_n \in D(A)$ for all n , such that $\lim_{n \rightarrow \infty} x_n = x \in X$ and $\lim_{n \rightarrow \infty} A(x_n) = y \in X$, then $x \in D(A)$ and $y = A(x)$.

We can characterize closable operators in terms of their graph:

Proposition 79. Let $A : D(A) \subset X \rightarrow X$. The following statements are equivalent:

1. A is closable,
2. $\overline{G(A)}$ does not contain points of the form $(0, z)$, $z \neq 0$,
3. A admits closed extensions.

Proof. 1. $\iff$ 2. A is not closable if and only if there exist two sequences $\{x_n\}_{n \in \mathbb{N}}$ and $\{x'_n\}_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} x'_n = x$, but $\lim_{n \rightarrow \infty} A(x_n) = y$ and $\lim_{n \rightarrow \infty} A(x'_n) = y'$ with $y \neq y'$. By linearity this is tantamount to saying that the sequence whose elements are $\tilde{x}_n \doteq x_n - x'_n$ converges to 0 as $n \rightarrow \infty$, while the sequence $A(\tilde{x}_n)$ converges to $y - y' = z \neq 0$. In turn, this amounts to saying that $\overline{G(A)}$ contains points of the form $(0, z)$ with $z \neq 0$. By negating this statement, the conclusion follows. It is also clear that the argument can be traced back to prove the reverse implication.

1. $\iff$ 3. If A is closable than $\bar{A}$ is a closed extension. Conversely, if A admits a closed extension, say B , per Definition 77 and 78, there cannot be in $\overline{G(A)}$ elements of the form $(0, z)$ with $z \neq 0$ since, otherwise, $(0, z) \in \overline{G(A)} \subset \overline{G(B)} = G(B)$. This is a consequence of $A \subset B$ and of B , being closed. Yet, this is also a contradiction since it would entail that, at the same time, the closed operator B is linear and $B(0) = z \neq 0$. $\square$

The situation improves if we assume that X is complete:

Lemma 80. *Let X, Y be Banach spaces, let $T \in \mathcal{B}(X, Y)$ and let $A : D(A) \subset Y \rightarrow Y$ be an operator on Y . If A is closable and $\text{Ran}(T) \subset D(A)$, then $AT \in \mathcal{B}(X, Y)$.*

Proof. Since A is closable, $AT = \bar{A}T$ on $D(A)$ and $\text{Ran}(T) \subset D(A)$. We can show that $\bar{A}T$ is closed. Consider a sequence $\{x_n\}_{n \in \mathbb{N}}$ such that both $\lim_{n \rightarrow \infty} x_n = x$, $x \in X$ and $\lim_{n \rightarrow \infty} \bar{A}T(x_n) = y$, $y \in Y$. Proposition 24 guarantees us that T is continuous, hence there must exist $z \in Y$ such that $\lim_{n \rightarrow \infty} T(x_n) = z$. Since $\bar{A}$ is closed per assumption, $z \in D(\bar{A})$ and $\bar{A}z = y$. Hence $(\bar{A}T)(x) = y$, which implies that $\bar{A}T$ is closed. We can now invoke the closed graph theorem 30 to conclude the proof. $\square$

One last remark is in due course. We have considered only Banach spaces since we know that the direct sum of two of them is still a Banach space, if endowed with the product topology. If we were to consider $\mathcal{H}, \mathcal{H}'$, two Hilbert spaces with Hermitian inner product $(\cdot, \cdot)$ and $(\cdot, \cdot)'$ respectively, we could transfer all results that we have obtained. In addition we also know that $\mathcal{H} \oplus \mathcal{H}'$ is again a Hilbert space if endowed with the Hermitian inner product

$$((x, y), (x', y'))_{\mathcal{H} \oplus \mathcal{H}'} \doteq (x, x') + (y, y'). \quad \forall (x, y), (x', y') \in \mathcal{H} \oplus \mathcal{H}' \quad (15)$$

We conclude this short section with a simple exercise

Exercise 16. Let $\mathcal{H}, \mathcal{H}'$ be two Hilbert spaces with Hermitian inner product $(\cdot, \cdot)$ and $(\cdot, \cdot)'$. Prove that $\mathcal{H} \oplus \mathcal{H}'$ is a Hilbert space (note that only the property of being complete is non trivial) and that the inner product (15) induces the product topology.

4.1 The Hermitian Adjoint Operator

As we have seen in the previous section, one of the key operations on bounded operators is that of adjunction. On the one hand, the adjoint operator exists as a by-product of Riesz's theorem,

while, on the other hand, one can identify the special class of self-adjoint operator. These are noteworthy since their eigenvalues are all real and, thus, they are the natural candidates for playing a pivotal role in quantum theories.

Unfortunately, the moment we drop the hypothesis of boundedness, we are back to square one since already Riesz's theorem cannot be used. We analyse now an alternative approach. Unless stated otherwise, we consider always a linear operator $T : D(T) \subset \mathcal{H} \rightarrow \mathcal{H}$ with a *dense* domain $D(T)$ in a Hilbert space $\mathcal{H}$. In principle we do not need to assume that it is separable, but it will not hurt doing so.

We recall that our guiding principle is looking for an operator T^* together with its domain of definition $D(T^*)$ so that the relation $(\phi, T\psi) = (T^*\phi, \psi)$ holds true for all $\psi \in D(T)$ and for all $\phi \in D(T^*)$. Bearing in mind the construction of the adjoint for bounded operators, we start from the following ansatz as far as the domain of the adjoint is concerned:

$$D(T^*) = \{\phi \in \mathcal{H} \mid \exists \Psi_\phi \in \mathcal{H} \text{ for which } (\phi, T\psi) = (\Psi_\phi, \psi), \forall \psi \in D(T)\}. \quad (16)$$

We observe that the definition is meaningful for the following reasons:

- $D(T^*)$ is never empty since at least $0 \in D(T^*)$ by setting $\Psi_0 = 0$. Furthermore it is a linear space since this property is inherited from the one of the Hermitian inner product,
- In general the assignment $D(T^*) \ni \phi \mapsto \Psi_\phi \in \mathcal{H}$ defines a linear functional if the image is unique. In general this is not the case unless $D(T)$ is dense. Suppose that, for a given $\phi \in D(T^*)$, there exists Ψ_ϕ and Ψ'_ϕ in $\mathcal{H}$ such that $(\Psi_\phi, \psi) = (\phi, T\psi) = (\Psi'_\phi, \psi)$, for all $\psi \in D(T)$. In other words $(\Psi_\phi - \Psi'_\phi, \psi) = 0$ if $\psi \in D(T)$. This identity yields no contradiction unless $D(T)$ is dense in $\mathcal{H}$. In this case, we consider a sequence $\{\psi_n\}_{n \in \mathbb{N}}$ such that $\psi_n \rightarrow \Psi_\phi - \Psi'_\phi$ as $n \rightarrow \infty$. Since, for all $n \in \mathbb{N}$, $(\Psi_\phi - \Psi'_\phi, \psi_n) = 0$ and since the Hermitian inner product is continuous, we can take the limit as $n \rightarrow \infty$, to obtain $\|\Psi_\phi - \Psi'_\phi\|^2 = 0$. This entails $\Psi_\phi - \Psi'_\phi = 0$.

These observations justify the following line of reasoning:

Definition 81. Let $\mathcal{H}$ be an Hilbert space and let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be a dense operator thereon. We call **adjoint** the operator $T^* : D(T^*) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ whose domain is (16) and whose image is $T^*(\phi) = \Psi_\phi \in \mathcal{H}$ for all $\phi \in D(T^*)$.

Notice that, even though T is densely defined there is no reason why T^* should also enjoy this property. Hence a priori $(T^*)^*$ might not be defined. On the bright side the adjoint enjoys some notable properties which are an almost straightforward consequence of the definition:

Exercise 17. Let T, T' be two densely defined operators on $\mathcal{H}$. Prove that

1. If $T \subset T'$ – see Definition 77, then $(T')^* \subset T^*$,
2. $(T')^*T^* \subseteq (TT')^*$ where the equality holds if $T \in \mathcal{B}(\mathcal{H})$.

Theorem 82. Let T be a densely defined operator on a Hilbert space $\mathcal{H}$. Then

1. T^* is closed and $G(T^*) = [\tau(G(T))]^\perp$ where

$$\tau : \mathcal{H} \oplus \mathcal{H} \rightarrow \mathcal{H} \oplus \mathcal{H}, \quad (\phi, \psi) \mapsto \tau((\phi, \psi)) = (-\psi, \phi). \quad (17)$$

2. T is closable if and only if $D(T^*)$ is dense and, in this case,

$$T \subseteq \overline{T} = (T^*)^*.$$

Proof. 1. Using (17), it holds

$$[\tau(G(T))]^\perp = \{(\phi, \psi) \in \mathcal{H} \oplus \mathcal{H} \mid -(\phi, T(\tilde{\psi}))_{\mathcal{H}} + (\psi, \tilde{\psi})_{\mathcal{H}} = 0, \forall \tilde{\psi} \in D(T)\}.$$

In other words we can identify ψ with $T^*(\phi)$, which means that $[\tau(G(T))]^\perp$ coincides with $G(T^*)$. In addition, since $[\tau(G(T))]^\perp$ is the orthogonal complement of a set in a Hilbert space, this is always closed. Hence the graph of T^* is closed and so is, per definition, the operator itself.

2. Let T be as according to the hypothesis and, by the theory of Hilbert spaces, it holds $\overline{G(T)} = [[G(T)]^\perp]^\perp$. In addition, since $\tau \circ \tau = -\mathbb{I}$, where $\mathbb{I}$ is here the identity operator on $\mathcal{H} \oplus \mathcal{H}$, and since, for any set $W \in \mathcal{H}$, $W^\perp = (-W)^\perp = -W^\perp$, we can conclude

$$\overline{G(T)} = -\tau \left(\left[\tau \left([G(T)]^\perp \right) \right]^\perp \right) = -\tau \left[G(T^*)^\perp \right] = \tau \left[G(T^*)^\perp \right],$$

where we used at different stages the results of item 1. and that τ commutes with the operation $\perp$ of taking the orthogonal complement. Proposition 79 guarantees us that $\overline{G(T)}$ is the graph of an operator, $\overline{T}$, if and only if it does not contain points of the form $(0, \psi) \in \mathcal{H} \oplus \mathcal{H}$. The above chain of identities tells us that this is equivalent to saying that there cannot exist $\psi \in \mathcal{H}$ such that $(0, \psi) \perp (-T^*\phi, \phi)$ for any $\phi \in D(T^*)$. Yet this holds true if and only if $\psi \in [D(T^*)]^\perp$. In other words $\overline{G(T)}$ is the graph of an operator if and only if the orthogonal complement of $D(T^*)$ is trivial, which is equivalent to saying that T^* is densely defined.

If this is the case, then $(T^*)^*$ is well-defined and, by item 1. of this theorem,

$$\overline{G(T)} = \tau \left[(G(T)^*)^\perp \right] = G((T^*)^*).$$

To conclude it suffices to observe that, per definition of closure $\overline{G(T)} = G(\overline{T})$. $\square$

Corollary 83. *Let T be a densely defined operator on a Hilbert space $\mathcal{H}$. Then*

1. $\ker(T^*) = [\text{Ran}(T)]^\perp$ and $\ker(T) \subseteq [\text{Ran}(T^*)]^\perp$ where the equality holds if $D(T^*)$ is dense in $\mathcal{H}$ and T is closed

2. if T is closed, then

$$\mathcal{H} \oplus \mathcal{H} = \tau(G(T)) \oplus G(T^*).$$

Proof. 1. This descends from the identity $(T^*\phi, \psi) = (\phi, T\psi)$ for all $(\phi, \psi) \in D(T^*) \oplus D(T)$, together with item 2. in Theorem 82.

2. If T is closed, then $\tau(G(T))$ is closed since τ is a unitary operator on $\mathcal{H} \oplus \mathcal{H}$. The statement descends from item 1. in Theorem 82 together with the standard properties of the orthogonal complement studied in the course of mathematical methods for physicists. $\square$

Exercise 18. Let T be a densely defined operator on a Hilbert space $\mathcal{H}$. Prove that

$$\ker(T^* - \lambda\mathbb{I}) = [\text{Ran}(T - \bar{\lambda}\mathbb{I})]^\perp, \quad \text{and} \quad \text{Ker}(T - \bar{\lambda}\mathbb{I}) \subseteq [\text{Ran}(T^* - \bar{\lambda}\mathbb{I})]^\perp,$$

where $\lambda \in \mathbb{C}$ and $\mathbb{I}$ is the identity operator on $\mathcal{H}$.

4.2 Essentially Self-Adjoint Operators

In view of the analysis of the previous part of this section, we are able to extend the notion of self-adjointness to operators which are not necessarily everywhere defined. More importantly we will be able to enlarge the class of potential physical observables by identifying a class of operators which are not self-adjoint, but admit a *unique* extension with such property. As we will emphasize, uniqueness is the key point and, in many cases, this is not available. The prime example will be the Laplace operator in a bounded region, say a box in $\mathbb{R}^n$, which admits multiple self-adjoint extensions, each of which can be interpreted as the choice of a boundary condition (Dirichlet, Neumann or Robin). We start with some nomenclature.

Definition 84. Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subset \mathcal{H} \rightarrow \mathcal{H}$. We say that T is

- **Hermitian** if $(\phi, T\psi) = (T\phi, \psi)$ for all $\phi, \psi \in D(T)$,
- **symmetric** if T is Hermitian and $D(T)$ is dense in $\mathcal{H}$,
- **self-adjoint** if $D(T)$ is dense and $T = T^*$,
- **essentially self-adjoint** if both $D(T)$ and $D(T^*)$ are dense and $T^* = (T^*)^*$.
- **normal** if $T^*T = TT^*$ where both sides are defined on the standard domain.

Observe that we have defined a new class of operators, those essentially self-adjoint which has no counterpart in the previous sections. In addition there are different, yet equivalent definitions of the above classes of operators. What a better occasion to offer you a new exercise? Actually, while reading this script, I realized that I am offering two exercises! This must be your lucky day (think how an unlucky day could turn out to be!)

Exercise 19. Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subset \mathcal{H} \rightarrow \mathcal{H}$. Prove that T is symmetric if and only if $\overline{D(T)} = \mathcal{H}$ and $T \subset T^*$.

Exercise 20. Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subset \mathcal{H} \rightarrow \mathcal{H}$. Prove that T is essentially self-adjoint if and only if $D(T)$ is dense in $\mathcal{H}$, T is closable and $T^* = \bar{T}$.

Before proceeding to a deeper analysis of essentially self-adjoint operators, we are falling short of one important answer, namely why we are calling the whole section "*unbounded operators*" when, up to now, we have been discussing mainly operators which are not everywhere defined. The following celebrated theorem gives (partly) a reason:

Theorem 85 (Hellinger-Toeplitz). *Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be an operator. If T is Hermitian and $D(T) = \mathcal{H}$ then it is bounded and self-adjoint.*

Proof. This is a consequence of item 1. of Proposition 39. □

Exercise 21. Let $\mathcal{H}, \mathcal{H}'$ be two Hilbert spaces and let $U : \mathcal{H} \rightarrow \mathcal{H}'$ be a unitary operator. If $T : D(T) \rightarrow \mathcal{H}$ and if $T' : D(T') \rightarrow \mathcal{H}'$ is such that

$$T' = UTU^{-1} \quad \text{and} \quad D(T') = UD(T),$$

then T' enjoys any of the following properties, if enjoyed also by T : closable, closed, Hermitian, symmetric, normal, self-adjoint and essentially self-adjoint.

IMPORTANT: Since we will be acting with the adjoint map several times on each operator, we shall adopt the notation T^{**} in place of $(T^*)^*$ and so on and so forth.

We start identifying criteria to distinguish essentially self-adjoint operators and to unveil their properties:

Proposition 86. *Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be an operator. Then*

1. *if $D(T), D(T^*)$ and $D(T^{**})$ are dense, then*

$$T^* = \overline{T^*} = \overline{T^*}^* = T^{***}.$$

2. *T is essentially self-adjoint if and only if $\overline{T}$ is self-adjoint,*
3. *If T is self-adjoint, it is maximally symmetric, that is it does not admit a proper extension to a symmetric operator,*
4. *if T is essentially self-adjoint, it admits one and only one self-adjoint extension, namely $\overline{T}$ which coincides with T^* .*

Proof. 1. Since $D(T), D(T^*)$ and $D(T^{**})$ are dense per hypothesis, we know that T^*, T^{**} and T^{***} exist. In view of item 2. of Theorem 82, we can conclude that $\overline{T^*} = T^{***} = (T^{**})^* = \overline{T^*}$. Item 1. of Theorem 82 entails that $T^* = \overline{T^*}$.

2. Suppose that T is essentially self-adjoint. As shown in Exercise 20, this entails that $\overline{T} = T^*$ and thus $D(\overline{T}) = D(T^*)$. By taking the adjoint, we obtain $\overline{T^*} = T^{**}$, but item 2. of Theorem 82 guarantees us that $\overline{T} = T^{**}$. Hence $\overline{T} = \overline{T^*}$ and, since the domains are dense, $\overline{T}$ is self-adjoint. Conversely suppose that $\overline{T}$ is self-adjoint. Then there exists its adjoint and $\overline{T^*} = \overline{T}$,

which implies that $D(T)$, $D(T^*)$ and $D(T^{**})$ are dense. In view of item 1. $T^* = \overline{T^*} = \overline{T}^*$. Hence $T^* = \overline{T}$ and T is essentially self-adjoint.

3. Suppose that T is self-adjoint and that there exists a symmetric operator K such that $T \subseteq K$. By taking the adjoint, we obtain $K^* \subseteq T^*$. Yet, since K is symmetric $K \subseteq K^*$. We have obtained the following chain of inclusions

$$T \subseteq K \subseteq K^* \subseteq T^*,$$

which collapses to equalities since $T = T^*$ by the hypothesis that T is self-adjoint.

4. Since T is essentially self-adjoint, we know from Exercise 20 that $\overline{T}$ is a self-adjoint extension. Suppose there exists a second one K . Forcefully $T \subset K$, which entails by taking the adjoint that $K = K^* \subseteq T^*$. Repeating the operation once more we obtain $T^{**} \subseteq K^{**} = K$. Yet, per definition of an essentially self-adjoint operator $T^* = T^{**}$. Hence we obtain that $K = K^* \subset T^* = T^{**} \subseteq K^{**} = K$. In other words $K = T^{**}$, but $T^{**} = \overline{T}$ in view of item 2. of Theorem 82. $\square$

Another relevant characteristic of an essentially self-adjoint operator is the following:

Theorem 87. *Let $\mathcal{H}$ be a Hilbert space and let T be a symmetric operator thereon. The following statements are equivalent:*

1. T is self-adjoint,
2. T is closed and $\ker(T^* \pm i\mathbb{I}) = \{0\}$ where $\mathbb{I}$ is the identity operator on $\mathcal{H}$,
3. $\text{Ran}(T \pm i\mathbb{I}) = \mathcal{H}$.

Proof. 1. $\implies$ 2. Suppose $T = T^*$. By item 1. of Proposition 82 we know that T^* is closed, hence T is closed. Suppose that $\psi \in \ker(T^* \pm i\mathbb{I})$. Then

$$T^*\psi = T\psi = \mp i\psi \implies \mp i(\psi, \psi) = (\psi, T\psi) = (T\psi, \psi) = (\mp i\psi, \psi) = \pm i(\psi, \psi),$$

which can hold true only if $(\psi, \psi) = 0$, i.e., $\psi = 0$.

2. $\implies$ 3. On account of Exercise 18, we know that $\ker(T^* \pm i\mathbb{I}) = [\text{Ran}(T \mp i\mathbb{I})]^\perp = \{0\}$. In other words $\text{Ran}(T \mp i\mathbb{I})$ is dense in $\mathcal{H}$. Since T and thus $T \pm i\mathbb{I}$ are closed and since $D(T)$ is dense, for all $\psi \in \mathcal{H}$ we can find a sequence $\{\psi_n\}_{n \in \mathbb{N}}$, with $\psi_n \in D(T)$ for all $n \in \mathbb{N}$, such that $\lim_{n \rightarrow \infty} (T\psi_n \mp i\psi_n) = \psi$. Notice that, for all $n \in \mathbb{N}$,

$$\|(T \mp i\mathbb{I})\psi_n\|^2 = \|T(\psi_n)\|^2 + \|\psi_n\|^2 \geq \|\psi_n\|^2.$$

This entails that since the left hand side converges as $n \rightarrow \infty$, so it does the right hand side. In other words there exists $\tilde{\psi} \in \mathcal{H}$ such that $\lim_{n \rightarrow \infty} \psi_n = \tilde{\psi}$. Since $T \mp i\mathbb{I}$ is closed $(T \mp i\mathbb{I})\tilde{\psi} = \psi$, that is $\text{Ran}(T \pm i\mathbb{I}) = \overline{\text{Ran}(T \pm i\mathbb{I})} = \mathcal{H}$.

3. $\implies$ 1. Since $T \subset T^*$, we need to show that $D(T^*) \subset D(T)$. Let $\psi \in D(T^*)$. Since $\text{Ran}(T \pm i\mathbb{I}) = \mathcal{H}$, there must exist $\phi \in D(T)$ such that

$$(T \pm i\mathbb{I})\phi = (T^* \pm i\mathbb{I})\psi.$$

Yet, $T = T^*$ on $D(T)$ and, thus, $(T^* \pm i\mathbb{I})(\psi - \phi) = 0$. The hypothesis entails also that $\ker(T^* \pm i\mathbb{I}) = [\text{Ran}(T \mp i\mathbb{I})]^\perp = \{0\}$, which yields in turn $\psi = \phi$ with $\phi \in D(T)$ per assumption and thus $\psi \in D(T)$. $\square$

Theorem 88. *Let $\mathcal{H}$ be a Hilbert space and let T be a symmetric operator thereon. The following statements are equivalent:*

1. T is essentially self-adjoint,
2. $\ker(T^* \pm i\mathbb{I}) = \{0\}$ where $\mathbb{I}$ is the identity operator on $\mathcal{H}$,
3. $\overline{\text{Ran}(T \pm i\mathbb{I})} = \mathcal{H}$.

Proof. 1. $\iff$ 2. If T is essentially self-adjoint, then T^* is self-adjoint and closed. We can apply to it the second criterion in Theorem 87 to conclude that $\ker(T^{**} \pm i\mathbb{I}) = \{0\}$. Since $T^{**} = T^*$ the sought statement is proven.

Suppose now that $\ker(T^* \pm i\mathbb{I}) = \{0\}$. Since T is symmetric, we know that $T \subseteq T^*$ and, by taking the adjoint $T^{**} \subseteq T^*$. Hence $D(T^*)$ is dense, being such $D(T)$. Once more item 2. of Theorem 82 entails that $T \subseteq \overline{T} = T^{**}$, which yields $\overline{T} \subseteq T^*$. The first item of Proposition 86 guarantees as a consequence that $T^* = \overline{T}^*$. In other words we have shown that $\overline{T}$ is symmetric. We can now apply item 2. of Theorem 87 to conclude that $\overline{T}$ is self-adjoint, but item 2. of Proposition 86 tells us that this is equivalent to having T essentially self-adjoint.

2. $\iff$ 3. This is a direct consequence of $[\text{Ran}(T \pm i\mathbb{I})]^\perp = \ker(T^* \mp i\mathbb{I})$ together with

$$\overline{\text{Ran}(T \pm i\mathbb{I})} \oplus [\text{Ran}(T \pm i\mathbb{I})]^\perp = \mathcal{H}.$$

The theorem stands proven. $\square$

We conclude this subsection with a key definition:

Definition 89. *Let $\mathcal{H}$ be a Hilbert space and let T be a closable, densely defined operator on $\mathcal{H}$. A dense subspace $W \subset D(T)$ is called a **core** for T if*

$$\overline{T|_W} = \overline{T}.$$

Proposition 90. *Let $\mathcal{H}$ be a Hilbert space and let T be a self-adjoint operator thereon. A subspace W is a core for T if and only if $T|_W$ is essentially self-adjoint.*

Proof. If $T|_W$ is essentially self-adjoint, then it admits a unique self-adjoint extension which must coincide with its closure. Hence $T|_W$ is a core. Conversely if $T|_W$ is a core, the closure is self-adjoint since it coincides with T itself. $\square$

4.2.1 The Position and the Momentum Operator – Definitions

In quantum mechanics there are two key examples which rely on the theory that we have developed so far: the momentum and the position operator. We will omit any proof since, for the first one, we need to get acquainted with the notion of Fourier transform and, for both, we need to study more in detail the smooth and compactly supported functions. We will discuss it in the second part of the course, hence we will postpone the analysis especially of the momentum operator to that moment. (ah ah ah moment and momentum!!!... The author needs holidays!)

Let $\mathcal{H} = L^2(\mathbb{R}^n, d^n x)$ where $d^n x$ indicates that we are considering the Lebesgue measure. We will omit henceforth indicating the measure which is left understood. For every $i = 1, \dots, n$ define the **position operators**

$$X_i : D(X_i) \subset \mathcal{H} \rightarrow \mathcal{H}, \quad \psi \mapsto x_i \psi, \quad (18)$$

where x_i represents the multiplication by the i -th coordinate over $\mathbb{R}^n$ endowed with the standard Euclidean coordinate system. The domain of definition is

$$D(X_i) \doteq \{\psi \in L^2(\mathbb{R}^n) \mid x_i \psi \in L^2(\mathbb{R}^n)\}. \quad (19)$$

Proposition 91. *Each position operator X_i , $i = 1, \dots, n$ on $D(X_i) \subset L^2(\mathbb{R}^n)$ is self-adjoint and $\mathcal{D}(\mathbb{R}^n) \equiv C_0^\infty(\mathbb{R}^n)$ is a core for X_i .*

As already mentioned all proofs are postponed to later sections of these notes. To define the momentum operator, we need a preliminary definition:

Definition 92. *A function $f : \mathbb{R}^n \rightarrow \mathbb{C}$ is called **locally integrable** if $fg \in L^1(\mathbb{R}^n)$ for all $g \in C_0^\infty(\mathbb{R}^n)$. We call $h = w - \partial_{x_j} f$ the **weak derivative** of f along the direction x_j , if both f and h are locally integrable and*

$$\int_{\mathbb{R}^n} d^n x h(x) g(x) = - \int_{\mathbb{R}^n} d^n x f(x) \frac{\partial g}{\partial x_j}. \quad \forall g \in C_0^\infty(\mathbb{R}^n)$$

If we observe that any element of $L^2(\mathbb{R}^n)$ is also locally integrable, we can call accordingly **momentum operator**

$$P_j : D(P_j) \subset L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n), \quad \psi \mapsto -i w - \partial_{x_j} \psi \quad (20)$$

with domain of definition

$$D(P_j) \doteq \{\psi \in L^2(\mathbb{R}^n) \mid \exists w - \frac{\partial \psi}{\partial x_j} \in L^2(\mathbb{R}^n)\}. \quad (21)$$

Proposition 93. *Each momentum operator P_j , $j = 1, \dots, n$ on $D(P_j) \subset L^2(\mathbb{R}^n)$ is self-adjoint and $\mathcal{D}(\mathbb{R}^n) \equiv C_0^\infty(\mathbb{R}^n)$ is a core for X_i .*

4.3 The Deficiency Indices

Up to now we have investigated those operators admitting a unique self-adjoint extension. In many interesting cases, such as for example the Laplace operator on a bounded domain, we have to give up on the request of uniqueness. We need therefore to establish useful criteria to decide whether and how many self-adjoint extensions exist. To this avail the first step is the extension of the concept of isometry to unbounded operators and the definition of the so-called Cayley transform:

Definition 94. Let $\mathcal{H}$ be a Hilbert space and let $U : D(U) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be an operator. It is called an **isometry** if

$$(U\psi, U\psi') = (\psi, \psi'), \quad \forall \psi, \psi' \in D(U)$$

which is equivalent to asking $\|U(\psi)\| = \|\psi\|$ for all $\psi \in D(U)$.

We recall that the map $\mathbb{R} \ni x \mapsto \frac{x-i}{x+i}$ is a bijection to $\mathbb{S}^1 \setminus \{1\} \subset \mathbb{C}$. This inspires the notion of Cayley transform

$$V(T) = (T - i\mathbb{I})(T + i\mathbb{I})^{-1}, \quad (22)$$

where T is an operator on a Hilbert space $\mathcal{H}$. In general the Cayley transform is not even well-defined, but the following proposition clarifies a lot its usefulness:

Proposition 95. Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be a symmetric operator. Then

- $T + i\mathbb{I}$ is injective,
- the Cayley transform (22) is well defined,
- $V(T)$ is an isometry and $\text{Ran}(V) = \text{Ran}(T - i\mathbb{I})$.

Proof. Since T is a symmetric operator, for every $\psi \in D(T)$, it holds that

$$\|(T + i\mathbb{I})\psi\|^2 = \|(T - i\mathbb{I})\psi\|^2 = \|T(\psi)\|^2 + \|\psi\|^2 \geq \|\psi\|^2. \quad (23)$$

Suppose $\psi \in \ker(T \pm i\mathbb{I})$. Hence $\|(T \pm i\mathbb{I})\psi\|^2 = 0$ implies $\|\psi\|^2 = 0$, which is tantamount to saying $\psi = 0$. In other words both $T \pm i\mathbb{I}$ are injective. Hence V is well defined on $\text{Ran}(T + i\mathbb{I})$ where $(T + i\mathbb{I})^{-1}$ exists. Notice also that $T - i\mathbb{I} = T + i\mathbb{I} - 2i\mathbb{I}$ and the standard domain of definition is $D(T + i\mathbb{I}) \cap D(\mathbb{I}) = D(T + i\mathbb{I})$ since $D(\mathbb{I}) = \mathcal{H}$. The same holds true exchanging i with $-i$. We need only to show that $V(T)$ is an isometry. Yet, for any $\phi \in D(V)$, $\tilde{\psi} \doteq (T + i\mathbb{I})^{-1}\phi \in D(T - i\mathbb{I})$. Hence by (23), we can conclude

$$\|V(T)\phi\| = \|(T - i\mathbb{I})\tilde{\psi}\| = \|(T + i\mathbb{I})\tilde{\psi}\| = \|\phi\|,$$

which the sought result. □

More generally we can prove

Proposition 96. *Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be an operator thereon. If the Cayley transform (22) is well-defined, then*

1. $\mathbb{I} - V$ is injective,
2. $\text{Ran}(\mathbb{I} - V) = D(T)$ and $T = i(\mathbb{I} + V)(\mathbb{I} - V)^{-1}$.

Proof. We observe that (22) is defined on $D(V) = \{\psi \in \mathcal{H} \mid \exists \phi \in D(T) \text{ such that } \psi = (T + i\mathbb{I})\phi\}$. For all $\psi \in D(V)$, $V\psi = (T - i\mathbb{I})\phi$, from which it descends

$$(\mathbb{I} + V)\psi = 2T\phi, \quad \text{and} \quad (\mathbb{I} - V)\psi = 2i\phi.$$

It follows that if $\psi \in \ker(\mathbb{I} - V)$, then $\phi = 0$, but, since $\psi = (T + i\mathbb{I})\phi$, also $\psi = 0$. At the same time, since $\phi \in D(T)$, $\text{Ran}(\mathbb{I} - V) = D(T)$ and, combining together the above two equalities, we obtain $T = i(\mathbb{I} + V)(\mathbb{I} - V)^{-1}$. $\square$

The most important result is the following:

Theorem 97. *Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be an operator thereon. Then*

- (a) *if T is symmetric, then it is self-adjoint if and only if $V(T)$ is unitary on $\mathcal{H}$,*
- (b) *if $V : \mathcal{H} \rightarrow \mathcal{H}$ is a unitary operator on $\mathcal{H}$ and $\mathbb{I} - V$ is injective, then it is the Cayley transform of a self-adjoint operator on $\mathcal{H}$.*

Proof. (a) Suppose that $T = T^*$. Then item 1. of Theorem 87 guarantees us that $\text{Ran}(T \pm i\mathbb{I}) = \mathcal{H}$. Item 3. of Proposition 95 entails that on these domain $V(T)$ is an isometry. Since $D(V) = \mathcal{H}$, V is also surjective, hence unitary.

Conversely, suppose that $V(T)$ is a unitary operator. Proposition 95 guarantees us that $\text{Ran}(T \pm i\mathbb{I}) = \mathcal{H}$ and item 3. of Theorem 87 that $T = T^*$.

(b) It suffices to show that V is the Cayley transform of a symmetric operator. Then item 1. of this Theorem allows us to draw the conclusion, we want. Per assumption, the map $\mathbb{I} - V$ is bijective from $D(V) = \mathcal{H}$ to $\text{Ran}(\mathbb{I} - V)$. Let us define the auxiliary operator A such that

$$A\psi = i(\phi + V\phi), \quad \forall \psi \in \mathcal{H} \text{ such that } \psi = \phi - V\phi. \quad (24)$$

Since ϕ is unique the operator A is well defined. Suppose $\psi, \psi' \in D(A) = \text{Ran}(\mathbb{I} - V)$ with $\psi = \phi - V\phi$ and $\psi' = \phi' - V\phi'$, with $\phi, \phi' \in D(V)$. Since V is per hypothesis an isometry

$$(A\psi, \psi') = -i(\phi + V\phi, \phi' - V\phi') = -i(V\phi, \phi') + i(\phi, V\phi') = (\phi - V\phi, i\phi' + iV\phi') = (\psi, A\psi').$$

In other words A is Hermitian. To show that A is symmetric, we must prove that its domain is dense. Yet $D(A) = \text{Ran}(\mathbb{I} - V)$ and $[\text{Ran}(\mathbb{I} - V)]^\perp = \ker(\mathbb{I} - V^*)$. The latter must be trivial, since, if there would exists $0 \neq \tilde{\psi} \in \ker(\mathbb{I} - V^*)$, then $V^*\tilde{\psi} = \tilde{\psi}$. Applying V and Exercise 11,

we obtain $V\tilde{\psi} = VV^*\tilde{\psi} = \tilde{\psi}$, namely $\tilde{\psi} \in \ker(V - \mathbb{I})$ which is impossible since $V - \mathbb{I}$ is injective. In other words $\text{Ran}(\mathbb{I} - V)$ and thus $D(A)$ are densely defined.

To conclude, we show that V is the Cayley transform of A as in (24). Notice that this equation entails that, for all $\psi \in \mathcal{H}$,

$$2iV\phi = A\psi - i\psi \quad \text{and} \quad 2i\phi = A\psi + i\psi$$

where $\phi \in D(A)$. As a by-product we observe that, since $\mathcal{H} = D(V) = \text{Ran}(A + i\mathbb{I})$, we have that

$$V(A + i\mathbb{I}) = (A - i\mathbb{I}),$$

namely V is the Cayley transform of A , as sought. $\square$

All these theorems suggest that the following objects play a key role:

Definition 98. Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be a symmetric operator thereon. Then we call **deficiency indices** of T

$$d_{\pm}(T) \doteq \dim(\ker(T^* \pm i\mathbb{I})). \quad (25)$$

We state now a theorem unveiling the key application of deficiency indices. Yet, since its proof is very long and technical, we omit it – the brave ones could refer to [5, Th. 5.36]

Theorem 99. Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be an operator thereon. Then

1. T admits self-adjoint extensions if and only if $d_+(T) = d_-(T)$.
2. If $d_+(T) = d_-(T)$, there is a bijection between the self-adjoint extensions of T and the surjective isometries from $\ker(T^* - i\mathbb{I})$ to $\ker(T^* + i\mathbb{I})$.

Proof. 1. Assume T admits a self-adjoint extension S , i.e. $T \subset S = S^*$. If we call $V(T)$ and $V(S)$ the Cayley transform of T and S respectively, item b) of Theorem 97 entails that $V(S) : \mathcal{H} \rightarrow \mathcal{H}$ is a unitary operator. In addition (22) guarantees that $V(T) \subset V(S)$. This is a by product of the fact that, since $T \subset S$, then $T - i\mathbb{I} \subset S - i\mathbb{I}$ and $(T + i\mathbb{I})^{-1} \subset (S + i\mathbb{I})^{-1}$. This entails also that $V(S)[\text{Ran}(T + i\mathbb{I})] \subset \text{Ran}(T - i\mathbb{I})$. Since $V(S)$ is a unitary operator, if $\psi \in \mathcal{H}$ is such that $\psi \in \text{Ran}(T + i\mathbb{I})^{\perp}$ then $V(S)\psi \in [(V(S)(\text{Ran}(T + i\mathbb{I})))]^{\perp} = \text{Ran}(T - i\mathbb{I})^{\perp}$. Here equality holds true since $V(S)$ is surjective, being a unitary operator. Using Exercise 18 we know that $\text{Ran}(T \pm i\mathbb{I})^{\perp} = \ker(T^* \mp i\mathbb{I})$ and thus $[(V(S)(\text{Ran}(T + i\mathbb{I})))]^{\perp} = \text{Ran}(T - i\mathbb{I})^{\perp}$ translates to $V(S)[\ker(T^* - i\mathbb{I})] = \ker(T^* + i\mathbb{I})$. This entails that the deficiency subspaces are in 1:1 correspondence, that is $d_+(T) = d_-(T)$.

Suppose now that T has equal deficiency indices and let $U_0 : \ker(T^* + i\mathbb{I}) \rightarrow \ker(T^* - i\mathbb{I})$ be a unitary operator. Using once more Exercise 18, we can decompose $\mathcal{H} = \overline{\text{Ran}(T + i\mathbb{I})} \oplus \ker(T^* - i\mathbb{I}) = \overline{\text{Ran}(T - i\mathbb{I})} \oplus \ker(T^* + i\mathbb{I})$. As a consequence we can construct a unitary operator $U : \mathcal{H} \rightarrow \mathcal{H}$ such that $U = V(T) \oplus U_0$ where $V(T)$ is the Cayley transform of T and where, for

every $\psi \in \mathcal{H}$, $U\psi = V(T)\phi + U_0\psi_0$ with $\psi = \phi + \psi_0$, $\phi \in \overline{\text{Ran}(T + i\mathbb{I})}$ and $\psi_0 \in \ker(T^* - i\mathbb{I})$. Observe that, since $V(T)$ is an isometry on $\text{Ran}(T + i\mathbb{I})$, it can be extended uniquely to an isometry on $\overline{\text{Ran}(T + i\mathbb{I})}$, cf. Proposition 26. It is worth mentioning that U is a bijection since injectivity is inherited from that of $V(T)$ and of U_0 , while surjectivity is a consequence of U_0 being bijective per assumption, while the (extension of the) Cayley transform of a symmetric operator establishes a 1:1 correspondence between $\overline{\text{Ran}(T - i\mathbb{I})}$ and $\overline{\text{Ran}(T + i\mathbb{I})}$. The operator $\mathbb{I} - U$ is injective. As a matter of fact, suppose $\chi \in \ker(\mathbb{I} - U)$. Writing $\chi = \chi_1 + \chi_0$ with $\chi_1 \in \overline{\text{Ran}(T + i\mathbb{I})}$ and $\chi_0 \in \ker(T^* - i\mathbb{I})$, $(\mathbb{I} - U)\chi = 0$ is equivalent to $V(T)\chi_1 = \chi_1$ and $\chi_0 = U_0\chi_0$. The second identity entails $\chi_0 \in \ker(T^* + i\mathbb{I}) \cap \ker(T^* - i\mathbb{I}) = \{0\}$. The first one can be solved only by $\chi_1 = 0$ on account of item 1. of Proposition 96. We can apply item b) of Theorem 97 to conclude that U is the Cayley transform of a self-adjoint operator T_{U_0} . Since $V(T) \subset U$ per construction, $T \subset T_{U_0}$. The number of self-adjoint extensions of T is ruled by the possible choices for U_0 .

2. We have already proven that for each surjective isometry between the deficiency subspaces there exists a self-adjoint extension of T . We need to prove the converse. Let S be such that $T \subset S = S^*$ and let $V(T)$ and $V(S)$ be the respective Cayley transforms. Per construction $V(T) \subset V(S)$ and $V(S)$ is a unitary operator on $\mathcal{H}$. Since $V(T)$ is a surjective isometry from $\overline{\text{Ran}(T - i\mathbb{I})}$ to $\overline{\text{Ran}(T + i\mathbb{I})}$ and since $\mathcal{H} = \overline{\text{Ran}(T + i\mathbb{I})} \oplus \ker(T^* - i\mathbb{I}) = \overline{\text{Ran}(T - i\mathbb{I})} \oplus \ker(T^* + i\mathbb{I})$, $V(S)$ must coincide with $V(T)$ on the first factor of the direct sum. In addition $V(S)$ must identify a unitary operator from $\ker(T^* - i\mathbb{I})$ to $\ker(T^* + i\mathbb{I})$. Furthermore, if S' were a second self-adjoint extension of T , its Cayley transform $V(S')$ would differ from $V(S)$ only for the unitary operator between the deficiency subspaces. This concludes the proof. $\square$

To conclude the section, we prove one last result:

Theorem 100. *Let $\mathcal{H}$ be a Hilbert space and let $T : D(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be an operator thereon. If T is symmetric, it is essentially self-adjoint if and only if it admits a unique self-adjoint extension.*

Proof. Suppose T is essentially self-adjoint. Then it has a unique self-adjoint extension by item 4. of Proposition 86. Theorem 99 guarantees instead that a symmetric operator has a self-adjoint extension only if it has the same deficiency indices, i.e., $d_+ = d_-$. Uniqueness is obtained if $d_+ = d_- = 0$. Yet the first point of Theorem 88 entails that A must be essentially self-adjoint. $\square$

To conclude the section, we explain how to use Theorem 99 to construct explicitly a self-adjoint extension of a symmetric operator such that $d_+(T) = d_-(T)$. Let us consider thus a surjective isometry $U : \ker(T^* - i\mathbb{I}) \rightarrow \ker(T^* + i\mathbb{I})$, then we indicate the extension of the operator T as $T_U : D(T_U) \rightarrow \mathcal{H}$ where

$$D(T_U) \doteq \{\chi \in \mathcal{H} \mid \exists! \phi \in D(\overline{T}) \text{ and } \psi \in \ker(T^* - i\mathbb{I}) \text{ such that } \chi = \phi + (\mathbb{I} + U)\psi\}, \quad (26)$$

from which we can infer that

$$T_U\chi = T\phi + i(\mathbb{I} - U)\psi. \quad \forall \chi \in D(T_U) \quad (27)$$